

claude-windows / 662b86ab-d2e3-4c02-8a0d-ec7a9071aba8

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\662b86ab-d2e3-4c02-8a0d-ec7a9071aba8\subagents\workflows\wf_4849400f-9d3\agent-aabd1e64eb2194.jsonl`  
- SHA-256: `2407eda0c72b15c1f85ea85ac3b33dfa405690d8f480f8073d3d8c50df458498`  
- Source modified: `2026-06-21T09:27:53+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_4849400f-9d3`  
- session_id: `662b86ab-d2e3-4c02-8a0d-ec7a9071aba8`
```

Transcript

1. user (2026-06-21T09:24:52.599Z)

You are mapping OWNER-GATED decisions for the badminton-highlight-indexer ENGINE serving track (project doc docs/COMPUTE_DECOUPLED_SERVING/README.md). Context: M0-M4 are SHIPPED (the config/input/output/device SEAM layer: deployment.profile+presets, input_backend, output base, DeviceContext). NOW the team is on M5+. The owner wants to BATCH-DECIDE every gate so the maximum independent code path is unblocked. The discipline: ONE isolated PR per milestone off origin/master, default-OFF + golden/cross-OS-gated for any core-caller change, mocked tests for anything not runnable locally (no GPU/cloud here). KEY DISTINCTION you must draw for every gate: separate (a) CODE that CAN be written + merged NOW behind default-OFF with mocked/faked tests (no owner needed) from (b) the precise OWNER PREREQUISITE (real GPU run, GCP spend, counsel/consent review, real prices, OAuth creds, a design pick). For each gate return the structured fields. Ground every claim in real file:line or doc anchors ? read the code, do not guess. Be exhaustive within your area but do not stray outside it.

YOUR AREA: cross-track gates + housekeeping that sit OUTSIDE the M-series but block or free independent work.

READ: docs/NEXT_STEPS.md (the prioritized cross-track pending list), docs/DOC_STATUS.md (§4 doc-hygiene leftovers), docs/PER_VIDEO_WORKSPACE.md (the P0-P6 plan; P3 output/chunks_* de-hardcode is said to be SUBSUMED by the shipped M3 output-base ? verify and frame the reconcile/close as a small Claude-doable PR), and grep the docs for WASB-C3 (weight provenance), collector md5 export, the DO droplet claude-do-rally-test + DO token retirement, and the repo rename off "badminton".

For each: classify housekeeping (Claude-doable cleanup/doc PR, NO owner needed) vs owner-action (retire droplet, rotate token, rename repo, counsel). The P3 reconcile and the Ops-Agent-codify-into-deploy/gcp item are likely Claude-doable PRs ? confirm. Report each as a gate with the right gate_type.

2. assistant (2026-06-21T09:25:01.009Z)

I'll map the cross-track gates and housekeeping in my area. Let me start by reading the key documents and grounding every claim in real file references.

3. user (2026-06-21T09:25:03.295Z)

Your configuration specifies to merge with the ref 'refs/heads/feat/serving-m4-device-context' from the remote, but no such ref was fetched.

4. user (2026-06-21T09:25:03.305Z)

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1      # Next steps (live) ? refreshed 2026-06-21  
2  
3      > **How to read:** this top section is the **current cross-track state + prioritized  
pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed  
**owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it  
is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc
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lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5    ## ? Where we are (2026-06-21)
6    **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only**
(ADR-013, DO dropped). **`COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ?
one pure core + local/cloud profiles; **owner M0 design sign-off pending**. **Per-video
workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273).
The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality /
owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
7
8    ## ? Prioritized pending ? all tracks
9    **P0 ? highest leverage / unblocks the most**
10   1. **[owner] `COMPUTE_DECOUPLED_SERVING` M0 sign-off** ? review the design + answer $8
gating Qs (cold-start, pricebook/currency, edge-auth, DPA/consent, quality-floor) ? unblocks M1.
11   2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3
matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus
training feed on it.
12   3. **[owner + Claude + GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the
`0.1.0-l4` bundle is n=2 plumbing-only).
13
14   **P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)**
15   4. **[Claude] `COMPUTE_DECOUPLED_SERVING` M1?M4** (config ? `input_backend` ? output-
base de-hardcode ? `DeviceContext`) once M0 is signed off.
16   5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 kills the
`output/chunks_` hardcodes ? P4 ? P5 Launch-Report flip ? P6 GPU parity).
17   6. **[Claude] Ops-Agent codification into `deploy/gcp`** (every VM auto-enables
observability ? [[gcp-ops-agent-enable]]).
18   7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
19   8. **[owner-side GPU] R-series: promote-or-reject `R2` toLF1 as the gate** (the
ablation) ? needs GPU-extracted golden features.
20
21   **P2 ? later / gated / deferred**
22   9. Rally-quality **P3/P4** (person-fusion LOV0, audio cue) ? needs GPU golden features.
23   10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
24   11. **Multisport** (table-tennis first, then pickleball).
25   12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within
`COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).
26   13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
27   14. **Rename repo off "badminton"** (owner, BACKLOG).
28   15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector
**`md5`** export (collector repo).
29   16. **PMF field validation** (owner-side, business track).
30
31   **Doc-hygiene (tracked in `docs/DOC_STATUS.md` $4):** ? #1 per-video reconcile + ? #2
this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary
confirm); confirm the COMPLETED top-level stubs; verify `HOSTING_PLAN`/`UI_PROPOSAL` currency.
32
33   ---
34
35   ## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)
36
37   **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
38   owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
39   ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
40   `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
**PR-2
41   upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
42   compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
43   coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into

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44     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
45     **Authoritative roadmap:**
46     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
47     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
48
49     ## ? Next-session / GPU-box pickup (2026-06-15)
50     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
51
52     **Immediate next step ? the real P3 LOVO measurement** (this box has a GPU +
`torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden
features + `.rallies.csv` labels are not yet here):
53     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
54     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
55     3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
56
57     **Also queued (owner-gated / pending input):**
58     - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
59     - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
60     - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
61
62     **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
63
64     ## Where we are (one paragraph)
65     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
66     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
67     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
68     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
69     sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
70     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
71     battery is done** (see table) ? the moat is quantified.
72
73     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
74     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
75     |---|---|---|
76     | Heuristic (velocity windowing) | 0.657 | 0.157 |
77     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
78     | Two-stage WASB+Gemini refine | 0.83 | ? |

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79 | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
80
81 **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
court drops the commodity
82 wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
pickups + dead-time merges,
83 the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
growth-sized gap, not an
84 architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
tapes (uploads candidate clips
85 only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
load-bearing.
86 Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
gemini_wrapper_2026-06-10.json).
87
88 ## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation
in progress)
89 **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR
#112 + session). All implementation is **owner-gated** (per explicit in the plan); each step =
ONE PR off `master`.
90
91 **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs
#156/#157/#160/#161/#163 + #165 + final records). ** See archived
`docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every
pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules
+ STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive
done; clean slate.
92
93 - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion
layer) .
94 `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
95 `docs/ROORA_LABELING_GUIDELINES.md` (v8) . `docs/HOW_RALLY_DETECTION_WORKS.md`
(2-layer mental model).
96 - **Key finding (no re-investigation needed): ** the owner's "held-shuttle counted as a
rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already
implements the net-crossing **"Gate A"* (`apply_gate(..., min_crossings=2)`) that kills
service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no
`min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via
fusion_hybrid) was the single cheapest, highest-confidence quality win. ** (P2 FusionSegmenter
specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists
`net_crossings` + optional person features, optional served Gate A/B via config knobs;
adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest
finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in
with knob for safety, and the leaderboard revert note.)
97 - **Current status (2026-06-14): ** P1 (person cue extraction to reusable
`PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment
harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-
ups): ** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos;
eager-person-compute optimisation); then P4 audio via hit_detector.
98 - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-
OFF, promote only on measured LOVO): **
99 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served
path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served
neutral on production windowing per litmus ? kept opt-in with knob for safety.)
100 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant
(players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the
eager compute optimisation note/landing. Pure + harness (testable here).
101 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
102 - **Discipline (from EXPERIMENT_HARNESS.md): ** every new cue **flag-gated, default
OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit
0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
103 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the
heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden
features + `rallies.csv` labels, which are not yet present in this checkout.
104 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are

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exhausted; full details + citations in RALLY_QUALITY_RESEARCH.md

\$5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):

- 105 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as "the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."
- 106 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.
- 107 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).
- 108 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8x fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
- 109 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.
- 110 6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."
- 111
- 112 **Discipline reminder:** owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.
- 113
- 114 ## Prioritized next steps
- 115 1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
- 116 Priority order for new videos: *(a) multi-court badminton* (the target distribution AND where the ship target
- 117 lives), *(b) ?3 matches per new sport ? table tennis first* (WASB transfers at F1 0.909; pickleball labels are
- 118 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per
- 119 `VIDEO\_RECORDING\_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video:
- 120 label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.
- 121 The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.
- 122 2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court
- 123 reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93
- 124 (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with
- 125 paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
- 126 3. **M0.4 next increment (\$0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the
- 127 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the
- 128 LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free
- 129 max-pool, ~2.8x fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review
- 130 (`RALLY\_QUALITY\_RESEARCH.md` \$5.6) ? better fit for the cheap/local/fast mandate.
- 131 4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across
- 132 badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated
- 133 by C3 (provenance multiplies per sport).
- 134 5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
- 135 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies
- 136 (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:

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137     `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner
ratified the
138     physical-visit approach; reframed product-feedback-first per owner spec):**
139     `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact
placeholders and recruits) +
140     `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student
script, demo-after-Q6
141     rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
142     capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md`
(G/A/R anchors + **pre-registered
143     decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
144     6. **Housekeeping:** ~rotate the Gemini API key~ **DONE 2026-06-10** (owner-verified
working; confirm old-key
145     deletion in AI Studio); **rename repo off "badminton"** (owner action,
146     in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
147
148     ## Pointers / artifacts
149     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` . Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) .
150     Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` . Quality history:
151     `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
152     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
153     manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
154     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) . eval manifest
`output/human_lovo_manifest.json`
155     (gitignored) . durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` .
156     labels: `...\Collect\Golden Labelled\`.
157     - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
158     (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
159     with a MISLEADING "credits depleted" 429; ramp gradually.
160     - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
161     PLATFORM slice = async job model (single-tenant).
162

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5. assistant (2026-06-21T09:25:06.031Z)

Now let me read the other key documents in my area.

6. user (2026-06-21T09:25:07.039Z)

```

1     # Documentation Status & Archival Register
2
3     > **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-
diligence**. **Last full review:** 2026-06-21.
4     > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a
milestone / delivered / archived), and re-review at each clean-slate.
5     > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update
it as work finishes** to prove we're done, and (c) **never archive a foundational doc without
honestly updating it first** (owner directive, 2026-06-21).
6
7     ---
8
9     ## 0. How to use this
10    - **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention.
§3 is the full register.
11    - **Marking something complete?** Update the doc's own **§7 progress tracker** (the
source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2
archival checklist.

```

12 - ****This register does NOT duplicate completeness detail**** ? it indexes lifecycle +
archive-readiness and points to each tracked doc's §7.

13

14 ## 1. Lifecycle legend

15 `DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: ****LIVING**** (permanent
reference, never "done"), ****PROPOSAL/DESIGN**** (owner-gated, not started), ****REPORT**** (a terminal
record of a run/launch), ****CONVENTION**** (a standing rule/guide).

16

17 ## 2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git
mv`)*

18 Before a foundational/tracked doc moves to `docs/archives/`:

19 1. ****VERIFY claims against the code**** ? no stale assertions; tag load-bearing facts
`[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it
first.)

20 2. ****Flip Status ? DELIVERED / DONE / REJECTED + a completion note**** ? what shipped,
where (PRs, paths, artifacts), and what was ****deferred / carried forward****.

21 3. ****Update ALL inbound references**** ? `docs/README.md` index, `CLAUDE.md`,
`docs/NEXT\_STEPS.md`, and any cross-doc links ? the new path + status.

22 4. ****Keep the lineage**** ? ADR forward/back-links + "supersedes / superseded-by" so the
evolution stays traceable.

23 5. ****`git mv` the whole dir/file ? `docs/archives/past\_projects/`**** (terminal-state
ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this
in the doc's header).

24 6. ****Record the move**** ? flip the row here to §3e + a line in memory `current-state`.

25

26 > ****Worked example (the template):**** ****DECOUPLED_COMPUTE ? #270**** ? verified M0?M2+M3.5
delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward
WASB-C3/DPA/`md5`), reindexed `docs/README` + `past\_projects/README`, kept the ADR-010?014
lineage, then `git mv` to `docs/archives/past\_projects/DECOUPLED\_COMPUTE/`.

27

28 ## 3. The register

29

30 ### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the
completeness source-of-truth)

Doc	Status	Archive-ready?	Next action
`COMPUTE_DECOUPLED_SERVING/`	DRAFT (owner-gated) ? created #272	No (just started)	Owner design sign-off (M0 gate) ? M1 config PR
`RALLY_QUALITY_RESEARCH.md`	Foundation, **tracked & ACTIVE** (§7)	No	Drives the R-series; complete when §7.1 single quality number is attributed
`R2_EVAL_METRIC_DESIGN.md`	IMPLEMENTED (augmenting)	No (gates R4 + dual-eval; ablation pending)	Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster
`R1_MILESTONE_LAUNCH_REPORT.md`	LAUNCHED (#204) ? terminal record	Soon (with the R-cluster)	Keep with R-series for narrative; archive when the track wraps
`RALLY_DETECTION_GAP_CLOSURE.md`	DRAFT (owner-gated)	No	G1?G3 levers; part of the R/quality cluster
`PER_VIDEO_WORKSPACE.md`	DRAFT, IN PROGRESS (P0 = #264)	No	**? supersedes `PER_VIDEO_OUTPUT_LAYOUT.md` ? reconcile (finding #1) ** ; then P1?P6
`MULTI_SIGNAL_FUSION_PLAN.md`	P1 DONE; P2+ owner-gated	No	P3/P4 (learned fusion, audio) ? needs GPU golden features
`EXPERIMENT_HARNESS.md`	P1 IMPLEMENTED	No	A/B + promotion gate; supports the quality track
`PROMINENT_COURT_DETECTION.md`	DRAFT (owner-gated)	No	C-series; multi-track selection design
`EXECUTION_POLICY.md`	P1 SHIPPED; P3b next	No	Async job policy; reused by Cloud Run dispatch
`GOLDEN_REGRESSION_FIXTURES.md`	IN PROGRESS (owner-gated)	No	Golden eval fixtures
Data layer ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` . `GOLDEN_SET_PHASE1_VIDEOS` . `GOLDEN_VIDEOS` . `VIDEO_RECORDING_GUIDELINES` . `GOLDEN_DATA_SHARING`) LIVING (data, **no-ML**) n/a Relocated 2026-06-21 to the isolated data pillar; tracked via `data_pipeline/README.md`. Roora *vendor* docs archived ? `archives/roora-vendor/`.			
`DESKTOP_APP_PLAN.md`	DRAFT (owner-gated, nothing built)	No	P0 compute-

45

feasibility spike; relates to truly-local profile |

46 | `OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |

47 | `PERSONALIZATION\_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB v5 store landed) |

48 | `UI\_ALPHA\_EXECUTION\_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |

49 | `I18N\_PLAN.md` | Design (P0); ****in implementation**** (adapted) | No | Executed in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated |

50 | `POST\_166\_...RISK\_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |

51

52 **### 3b. Foundational / strategy references (LIVING ? archive only if ****superseded****, not when "done")**

53 | Doc | Status | Note |

54 |---|---|---|

55 | `PLATFORM\_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING |

56 | `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |

57 | `VIDEO\_LOCALITY\_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |

58 | `HOSTING\_PLAN.md` | PROPOSAL | ****? partly overtaken by ADR-013 + the shipped site ? verify (finding #5)**** |

59 | `UI\_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half ****in implementation**** ? `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc) |

60 | `STORAGE\_SHARING\_MODEL.md` | DRAFT (co-design pending) | ? |

61 | `ANALYZERS\_AND\_RECONCILERS.md` | DESIGN (owner-gated) | ? |

62 | `ROADMAP.md` / `CODE\_MAP.md` / `HOW\_RALLY\_DETECTION\_WORKS.md` / `KHELSUTRA\_PRODUCT\_OVERVIEW.md` | LIVING references | Keep current; not archive targets |

63

64 **### 3c. Reports & terminal records (archive candidates once their track wraps)**

65 | Doc | Status | Note |

66 |---|---|---|

67 | `REAL\_FOOTAGE\_VALIDATION\_REPORT.md` | validation complete | Motivates R2; archive with the R-cluster |

68 | `RALLY\_DETECTION\_QUALITY\_REPORT.md` | report | Quality snapshot; part of R-cluster |

69 | `QUALITY\_ITERATIONS.md` | LIVING empirical log | ****Keep**** ? the paper-backbone ablation log ([[paper-coauthor-aim]]) |

70 | `CODE\_AUDIT\_AND\_TEST\_HARDENING.md` · `HARDENING\_LOOP\_HANDOFF.md` · `DEFERRED\_HARDENING\_PLAN.md` | COMPLETED ? ****top-level pointer stubs**** (archived copies exist) | ****? confirm intentional stubs (per #165) vs finish-archiving (finding #4)**** |

71

72 **### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)**

73 `PROJECT\_DOC\_TEMPLATE.md` · `NEXT\_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION\_LAYER.md` · `SETUP\_NEW\_MACHINE.md` · `TRACKNET\_WSL\_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data\_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

74

75 **### 3e. Already archived (`docs/archives/`)**

76 `past\_projects/DECOUPLED\_COMPUTE/` (? 2026-06-21, #270) · `past\_projects/low-regret-moves/` · ****field-pmf/**** (field-ops + customer-feedback, 2026-06-21) · ****roora-vendor/**** (annotation-vendor engagement, 2026-06-21) · `PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE\_AUDIT\_AND\_TEST\_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED\_HARDENING\_PLAN-COMPLETED-2026-06-14.md` · `HARDENING\_LOOP\_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL\_READINESS\_REVIEW.md` · `MONETIZATION\_AUDIT.md` · `TEST\_RUN\_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/`.

77

78 **### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`**

79 ****ADR-001 ? ADR-014**** (14 ratified/proposed). The compute/serving lineage: ****ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014****. ADR-014 (PROPOSED) is the only non-ratified one (owner sign-off = the COMPUTE_DECOUPLED_SERVING M0 gate).

80

81 **## 4. Open due-diligence findings / actions (2026-06-21 review)**

82

83 > ****? Doc-restructuring done (2026-06-21, owner-directed):**** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML ****data**

layer** carved into `docs/data\_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data\_pipeline` one sets the "no segmentation/stitching" boundary).

84 1. **? RESOLVED (2026-06-21) ? `PER\_VIDEO\_WORKSPACE.md` supersedes
`PER\_VIDEO\_OUTPUT\_LAYOUT.md`.** Folded forward the `video\_slug` naming option (workspace §8.5) +
the `output/GX010125\_run/` manual-prototype precedent; predecessor archived ?
`docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE\_MAP.md` repointed.

85 2. **? RESOLVED (2026-06-21) ? `NEXT\_STEPS.md` refreshed** with a current 2026-06-21
state + a **cross-track prioritized pending list** ; the 2026-06-13 owned-model/quality body is
kept (clearly marked) as that track's detail.

86 3. **R-series / quality cluster is ACTIVE** (`R1`,`R2`,`RALLY\_DETECTION\_GAP\_CLOSURE`,`RA
LLY_QUALITY_RESEARCH`,`QUALITY_ITERATIONS`,`REAL_FOOTAGE_VALIDATION_REPORT`,`RALLY_DETECTION_QUA
LITY_REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected
as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is",
2026-06-21.)

87 4. **Confirm the COMPLETED top-level stubs** (`CODE\_AUDIT\_AND\_TEST\_HARDENING`,
`HARDENING\_LOOP\_HANDOFF`,`DEFERRED\_HARDENING\_PLAN`) are intentional pointer-stubs (per #165) ?
or finish-archive them so there's one copy.

88 5. **`HOSTING\_PLAN.md` / `UI\_PROPOSAL.md` are PROPOSAL but partly overtaken** by ADR-013
(DO dropped) + the shipped khelsutra site. **? `UI\_PROPOSAL.md` banner added (2026-06-21)** ?
its UX half is now tracked in `khelsutra/docs/FRIENDLY\_AND\_MULTILINGUAL.md`; the `I18N\_PLAN.md`
localization track is likewise in implementation there. **`HOSTING\_PLAN.md` banner still
pending** (ADR-013 already noted in it).

89 6. **`COMPUTE\_DECOUPLED\_SERVING` M0 sign-off pending** ? owner reviews the design +
answers §8 gating questions before M1.

90

91 ## 5. Maintenance

92 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at
each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md`
archive policy and [[working-agreement-tracked-project-docs]].

93

7. user (2026-06-21T09:25:07.249Z)

1 # Per-video workspace ? one folder per video, local?cloud identical

2

3 > **Status:** `DRAFT` ? `in progress` (owner-requested 2026-06-20) . **Owner:** Avi .
Created: 2026-06-20 . **Last updated:** 2026-06-21

4 > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)

5 > **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory
`current-state`.

6 > **Relation to existing docs:** peer-of `DECOUPLED\_COMPUTE` (reinforces the
storage/compute seam) and the appliance track; extends `file-placement-conventions`.
Supersedes `PER\_VIDEO\_OUTPUT\_LAYOUT.md` (the 2026-06-14 design predecessor ? archived
2026-06-21 as `docs/archives/PER\_VIDEO\_OUTPUT\_LAYOUT-SUPERSEDED-2026-06-21.md`). **Folded in
from it:** (a) the **`video\_slug` naming alternative** ? a human-readable folder name
`sanitized-stem + video\_id[:12]` (e.g. `GX010125\_89605e80ed01`) vs this doc's full-MD5
`video\_id` (D1) ? see open-Q §8.5; (b) the **`output/GX010125\_run/` manual prototype**
(2026-06-14) = the first hand-made per-video isolation, the precedent this automates.

7 > **Honesty note:** claims tagged `[verified]` (checked vs code by the mapping
workflow), `[design]` (proposed), `[researched]`.

8

9 ---

10

11 ## 0. TL;DR

12 Today a single video's artifacts are **scattered** ? `output/<family>/\*.csv`,
`output/chunks\_<provider>\_handoff/`,`~/clips/<key>\_frames/`,`<scratch>/outputs/<video\_id>/`,`
reels in `output/`,` logs to stdout, plus DB rows. The mapping workflow found **56 write sites
across 4 subsystems** `[verified]`. Goal: **one folder per video, keyed by the MD5 `video\_id`,
holding everything about that video** (working video copy/pointer, frames, trajectory,
candidates, intervals/report, reel, run log) **except its metadata, which stays in the DB**. The
root is a **storage URI**, so a **local** deployment and a **cloud (bucket)** deployment use the
same relative layout ? only the root differs ? easy to find, easy to repro, easy to clean.
Single most important caveat: this is a **default-OFF, phased** convergence (legacy flat layout
untouched until a Launch-Report flip) because the worker's bucket push prefix and the WSL/native

fast-scratch paths are load-bearing.

```
13
14     ## 1. Problem & goal
15     - **Why now:** approaching launch + the DECOUPLED/Cloud-Run serving track. The owner
wants per-video repro parity across local and cloud ("everything about a video resides in one
place; metadata in the DB"). Ties [[feedback-launch-quality-bar]] (serious observability) and
the Cloud-Run subproject (ADR-012).
16     - **Good looks like:** given a `video_id`, a single `ls <root>/<video_id>/` shows *all*
of its artifacts, identically whether `<root>` is a local dir or `gs://?`. Deleting that one
folder fully cleans the video (DB row aside).
17     - **Read to get current:** the §4 reuse map;
`backend/utils/hashing.py::compute_video_id`; `backend/storage/` (URI abstraction);
`backend/worker/__main__.py`; `backend/pipeline/segmenters/trajectory_hybrid.py`.
18
19     ## 2. Decisions locked
20     Defaults chosen to keep momentum; the ?-marked ones in §8 are still owner-confirmable.
21
22     | # | Decision | Source / date | Implication |
23     |---|-----|-----|-----|
24     | D1 | **Key = full MD5 `video_id`** for the folder name (`<root>/<video_id>/`). | owner
ask, 2026-06-20 | Matches `videos.id` / the storage join key; collision-proof. Artifact
*filenames* keep the existing `video_id[:12]` infix. |
25     | D2 | **Root is a storage URI** ; local & cloud share ONE relative layout (only the root
differs). | design, [[monetization-plan]] | Reuses `backend/storage/` dispatch
(`local://`/`gcs://`/`s3://`/`gdrive://`). No per-deployment code. |
26     | D3 | **Two roots:** a *durable* workspace root (may be `gs://`) for artifacts to keep;
a *fast local compute-scratch* mirror (always local) for frame caches/chunks. Compute writes
locally, then **pushes durable artifacts** to the workspace. | design + back-compat | WSL/native
frame extraction never writes to a cloud root (perf + correctness). |
27     | D4 | **DB stays metadata-only** + ONE additive nullable `videos.workspace_uri` (v6).
`NULL` = legacy flat layout. | design | No artifact bytes in the DB; the pointer makes a video's
folder discoverable. |
28     | D5 | **Default-OFF behind a flag** (`storage.single_folder_per_video`); converge in
phases; flip via a **Launch Report**. | [[launch-report-convention]] | Zero regression risk
during convergence; the deployed bucket layout only changes on the gated flip. |
29     | D6 | **NOT per-video (stay outside the folder):** the `weights/<version>/` bundle
(version-keyed), the SQLite DB, the WSL/native `wasb_infer.py` staging copy (infra), and the
training corpus (`labels/`, `videos/`) + eval fixtures (shared inputs). | [verified] mapping |
The convention covers *per-video* artifacts only; corpus/eval/model paths are intentionally
excluded. |
30
31     ## 3. Foundation ? the artifact map `[verified]`
32     The mapping workflow (`wf_085a5118`, 5 agents) catalogued every per-video write site.
Consolidated:
33
34     | Artifact | Today | Keyed by | Target subfolder |
35     |---|---|---|---|
36     | Working video copy (cloud fetch) / source | `scratch/in.mp4`, `~/clips/<key>.<ext>` |
video_id | `source/` (copy) or `source/source.uri` (local pointer) |
37     | Trajectory CSV (wasb/predict/stub) | `output/<family>/<key>*.csv` | video_id |
`detector/` |
38     | WASB detector cache (`det_raw.jsonl`, `manifest.json`) |
`<csv_dir>/<key>_wasb_wasbcache/` | video_id | `detector/<key>_wasbcache/` |
39     | Frame PNG cache | `~/clips/<key>_frames/`, `scratch/<key>_frames/` | video_id |
`tmp/` (**local-only**) |
40     | Handoff clips (AI validation) | `output/chunks_<provider>_handoff/` (**hardcoded**) |
video_id+idx | `tmp/` (**local-only**) |
41     | YOLO person chunks | `output/chunks_yolo/` (**hardcoded**) | video_id+idx | `tmp/`
(**local-only**) |
42     | Gemini chunks / temp-trim | `output/chunks/<video_id>/`,
`output/temp_trimmed_<video_id>.mp4` | video_id | `tmp/` (**local-only**) |
43     | `intervals.csv`, `report.json` (worker) | `scratch/outputs/<video_id>/` ?
`<out_uri>/outputs/<video_id>/` | video_id | `outputs/` |
44     | Highlight reel | `output/<basename>.mp4` | video_id (via job) | `reels/` |
45     | Per-segment compile clips | `tempdir/clip_<idx>.mp4` | seg idx | `tmp/` (**local-
```

```

only**) |
46 | Obsreport JSON | `output/<video_id>_report.json` | video_id | `outputs/` |
47 | Run log | stdout (not persisted) | ? | `logs/run.log` |
48 | Candidate/play segments, jobs, events | SQLite tables | video_id | **DB only**
(metadata) |
49
50 Two genuinely **hardcoded** dirs to kill: `output/chunks_<provider>_handoff` and
`output/chunks_yolo` (this subsumes the long-carried "de-hardcode `output`" item,
`trajectory_hybrid.py:237,313`).
51
52 ## 4. Design / architecture
53 **Target layout (per video):**
54 ```
55 <workspace_root>/<video_id>/
56   source/      # working copy (cloud-fetched) OR source.uri pointer (local)
57   detector/    # trajectory CSV(s) + <key>_wasbcache/ (det_raw.jsonl, manifest.json)
58   outputs/     # intervals.csv, report.json, <video_id>_report.json (obsreport)
59   reels/      # compiled highlight mp4(s)
60   logs/       # run.log (per-video processing log)
61   tmp/        # LOCAL-ONLY transient: frame PNGs, handoff/yolo/gemini chunks, compile
clips ? never pushed; cache-hygiene drops this
62   ```
63 **Outside the per-video folder** (D6): `weights/<version>/`, the SQLite DB,
`wasb_infer.py` staging, `labels/`+`videos/` corpus, eval fixtures.
64
65 **Reuse map (what's new vs reused):**
66 - **NEW** `backend/utils/workspace.py` ? pure path helper: `video_workspace(root,
video_id, *, sub=None, name=None)` builds `

```

```

86 | P4 | Converge **frame cache / WSL+native staging / reels / obsreport / run log** onto
`tmp/`+`logs/`+`reels/`+`outputs/`, flag-gated | P3 | No | ? | ? |
87 | P5 | **Launch Report** + flip default ON + retarget `tools/cleanup_caches.py` + docs |
P2,P3,P4 | ? owner | ? | ? |
88 | P6 | Validate the flag on a real GPU run (local + a `gcs://` workspace) ? repro parity
| P5 | ? owner GPU | ? | ? |
89
90 ## 8. Open questions ? owner *(blocking-gates vs nice-to-knows)*
91 **Proposed defaults applied (override if you disagree):**
92 1. **Source video: copy or pointer?** ? *default:* cloud-fetched copy lands in
`source/`; a local source is referenced by a `source/source.uri` pointer (no duplication of
large files). ? confirm.
93 2. **Cloud root namespace?** ? *default:*
`gs://<bucket>/<workspace_prefix>/<video_id>/` with `workspace_prefix="workspaces"` (keeps per-
video folders tidy beside `videos/`, `labels/`, `weights/`; the legacy `outputs/<video_id>/` stays
until P5). ? confirm (affects the deployed bucket contract).
94 3. **Push transient chunks/frames to cloud?** ? *default:* **No** ? `tmp/` is local-only
(huge + ephemeral). nice-to-know.
95 4. **Migration of existing artifacts?** ? *default:* **forward-only** (legacy flat
layout left in place; optional one-shot migrator later). nice-to-know.
96 5. **Folder name ? full MD5 vs human-readable slug?** ? *default (D1):* full MD5
`video_id` (collision-proof, matches the DB key). The predecessor `PER_VIDEO_OUTPUT_LAYOUT.md`
preferred a **`video_slug`** = `sanitized-stem + video_id[:12]` (e.g. `GX010125__89605e80ed01`)
for navigability. Trade-off: slug is human-readable but the stem can collide / be renamed; full-
MD5 is opaque but stable. nice-to-know (the helper can expose both).
97
98 ## 9. Definition of done
99 - [ ] Helper + config land with tests (P0).
100 - [ ] `videos.workspace_uri` migration (P1).
101 - [ ] All $3 per-video write sites route through the helper under the flag (P2?P4); the
two hardcoded `output/chunks_` dirs are gone.
102 - [ ] Launch Report shows no regression; flag flipped ON; cleanup retargeted (P5).
103 - [ ] One real GPU run proves a `<video_id>/` folder is complete + identical local vs
`gcs://` (P6).
104 - [ ] ? flip Status DONE, archive.
105
106 ## 10. References
107 **Internal code anchors `[verified]`** : `backend/utils/hashing.py`, `backend/storage/`,
`backend/worker/__main__.py:98,146-158,282-340`,
`backend/pipeline/segmenters/trajectory_hybrid.py:237,313`, `yolo_hybrid.py:268,407`,
`gemini.py:23,119`, `backend/pipeline/detectors/{wasb,native_wasb,tracknet,stub}_runner.py`,
`wasb_infer.py:90-94`, `backend/pipeline/stitcher/compiler.py:233,326`,
`backend/reporting/__init__.py:72`, `backend/infrastructure/database.py`. **Decisions:** ADR-012
(Cloud-Run serving). **Mapping artifact:** workflow `wf_085a5118` (56 write sites).
108
109 ---
110 ### Changelog
111 - 2026-06-20 ? created from the mapping-workflow output; P0 in progress.
112

```

8. assistant (2026-06-21T09:25:09.784Z)

Now let me read the M3 output-base milestone in the serving doc and verify the P3 subsumption claim, plus grep for the cross-track items.

9. user (2026-06-21T09:25:11.585Z)

```

1 # Compute-Decoupled Rally Serving ? one pure core, profile-selected compute + storage
2
3 > **Status:** ? **M0 APPROVED** (owner, 2026-06-21) ? **IN PROGRESS (build M1+)** .
**Owner:** Avi . **Created:** 2026-06-21 . **Last updated:** 2026-06-21
4 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to
`docs/archives/past_projects/` when DONE)
5 > **Tracking anchors:** $7 progress tracker is the source of truth; pointer in memory
`current-state` + `docs/NEXT_STEPS.md` once work starts.

```

6 > **Relation to existing docs:** the productionization successor to the archived
`DECOUPLED\_COMPUTE` (M3/M4); realizes the `ComputeBackend`/`StorageBackend` registries from
[PLATFORM_ARCHITECTURE.md](../PLATFORM_ARCHITECTURE.md) §3; the **truly-local** profile is the
L0 tier of [VIDEO_LOCALITY_MODEL.md](../VIDEO_LOCALITY_MODEL.md) and the `LocalDesktop` backend
of [DESKTOP_APP_PLAN.md](../DESKTOP_APP_PLAN.md).

7 > **Naming:** this was the "Cloud Run + GPU serving subproject" (ADR-012). Renamed
compute-decoupled serving because **truly-local** is a co-equal, first-class profile, not an
afterthought.

8 > **Honesty note:** claims tagged `[verified]` (checked against code by the design
workflow `wf\_56274ff0`), `[design]` (proposed), `[researched]`.

9

10 ---

11

12 > **? Northstar (D14):** a **mobile, app-like** flow ? point at a **Google Drive** video
? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive**, next
to the source. Two design promises around it: a user can **switch local?cloud anytime** (D13,
a headline feature), and execution is **never a surprise** ? auto-detect **suggests**, the user
confirms (D15).

13

14 ## 0. TL;DR

15 The rally engine is a **pure function** of `(input bytes + config) ? outputs` in three
deterministic stages ? **DETECT ? WINDOW ? STITCH** ? that **no deployment profile** may branch
on. A **ComputeBackend** (deployment profile) decides **where** the core runs (a local process
vs a Cloud Run GPU job); a **StorageBackend** decides **where bytes come from** (local FS / USB
/ mounted Google Drive / bucket). **One startup config** picks both. We ship **truly-local**
(local GPU + in-process stitch, zero cloud) and **cloud-serving** (upload `<2GB` / gdrive /
bucket ? a Cloud Run GPU job runs detect **and** stitch, so stitch's compute is metered), plus
cpu-mock for CI. The single most important caveat: detection is **byte-identical** only on the
same GPU (cross-GPU is sub-pixel-different, per the 2026-06-20 deploy report) ? so the
parity guarantee is enforced at the WINDOW + STITCH layer, not detector-CSV bytes.

16

17 ## 1. Problem & goal

18 - **Why now:** DECOUPLED_COMPUTE proved compute **portability** (M0?M2 + M3.5 on a GCP L4)
and is archived; its deferred **M3** (serving endpoint + per-video billing) + M4 (scale-to-zero +
cost guard) become this project. The owner needs **both** a hosted service **and** a truly-local
mode, with the **core** (detect+stitch) unaffected by which one runs.

19 - **The two user-facing outcomes** (owner, 2026-06-21):

20 - **(a) Hosted service** ? operate on small **local uploads** (`<2GB`) **and** load from
gdrive / a bucket, spinning up **Cloud Run** for both rally segmentation AND stitching
(stitching is compute-intensive ? run it where its cost is accounted for).

21 - **(b) Truly-local** ? on a local machine with videos on **local drives/USB** + a local
GPU, run inference **and** stitching **completely locally**.

22 - **"Good"** looks like: the same `video` ? rallies ? reel` core gives identical windows
+ stitched ranges whether it runs on a laptop or Cloud Run; switching is a **config/startup**
choice, never a code branch in the core.

23

24 ## 2. Decisions locked

25

#	Decision	Source	Implication
D1	The core is a pure 3-stage function (DETECT/WINDOW/STITCH); no profile branch inside it. design `[verified]` All profile difference lives in config + the two registries.		
D2	Two registries: `ComputeBackend` (where) selected by `deployment.profile` + `compute_target`; `StorageBackend` (what-bytes) by `input_backend`/`storage.backend`. ADR-014 Realizes PLATFORM_ARCHITECTURE's `local`/`hosted`/`byo_worker`.		
D3	Profiles shipped: `truly-local`, `cloud-serving`, `cpu-mock` (CI); `byo_worker` reserved/deferred. ADR-014 Enum/seam reserved so BYO isn't precluded.		
D4	In cloud-serving , STITCH runs on Cloud Run (co-located with detect) so CPU/wall + egress are metered into the per-job cost ledger. owner (a) Stitch-on-laptop would hide real cost.		
D5	Parity enforced at WINDOW+STITCH (byte-exact for a fixed trajectory); detect is byte-exact only same-GPU. deploy report 2026-06-20 Cross-GPU asserted at the rally/window level, never CSV bytes.		
D6	Default-OFF , smallest-safe-first, one PR per milestone; any core-caller		

change (e.g. configurable output base) ships behind a flag + a Launch Report. | [[launch-report-convention]] | Zero-regression discipline. |

34 | D7 | ****Cloud Run cold-start: SCALE-TO-ZERO**** ? accept the ~1?3 min cold-start (it's an async job: upload?poll?reel, amortized over the multi-minute job). ****No warm pool**** (that ~\$500/mo idle would break per-video billing). | owner 2026-06-21 | M6/M7. Revisit a warm lane only on a UX complaint. |

35 | D8 | ****Pricebook = a versioned DB table****; cost computed in ****USD**** (matches GCP billing = one source of truth), ****displayed in INR**** at a configurable FX rate (Bangalore market); re-versioned when GCP prices change. | owner 2026-06-21 | M8. |

36 | D9 | ****Edge auth = Cloudflare Access**** (reuse the site's existing CF). Lock the Cloud Run ****ingress to the CF proxy**** + ****verify the `Cf-Access` JWT signature**** (so a direct hit to the Cloud Run URL can't spoof the identity header). | owner 2026-06-21 | M9. One identity system. |

37 | D10 | ****Free-tier "data-for-reels" trade:**** free reels in exchange for ****training rights**** (uploaded footage + derived trajectories/labels); ****paid tier = no-train**** (privacy is the paid feature). ****Carve-outs:**** train ONLY on attested ****ADULT**** footage (minor footage ? reel yes, training pool ****NO**** ? DPDP/GDPR need verifiable parental consent); train on ****DERIVED**** data + ****auto-delete raw****; ****ToS counsel-reviewed****; live training-on-uploads ****gated to M9**** (public signup). Truly-local needs no DPA. | owner 2026-06-21 | M9 + D12; ties [[paper-coauthor-aim]] / PERSONALIZATION flywheel. |

38 | D11 | ****Quality floor: Gemini handoff ON**** for cloud-serving until the owned model clears the #172 ship-gate (e.g. ?0.70 multi-court) ? then flip via a ****Launch Report****; the owned model runs in ****shadow/eval**** meanwhile. (Gemini = serving/inference ONLY, never a training source ? CLAUDE.md guardrail.) | owner 2026-06-21 | M7 + D12. |

39 | D12 | ****Data-flywheel harness is IN SCOPE** (owner add, Q5): ****capture the (consented, adult) uploaded video + the Gemini reel/rally output ? **reliably store**** in the per-video workspace/bucket ? ****run shadow evals**** (owned-vs-Gemini, dual-eval-set) ? ****promote good ones to the golden TRAINING set**** (human-reviewed labels) so the owned model climbs toward the D11 floor (then Gemini flips off). | owner 2026-06-21 | new ****M11****; ties D10 (consent) + `data\_pipeline/` + the eval/experiment harness. |

40 | D13 | ****Profile is user-switchable ANYTIME ? a headline FEATURE.**** The same user runs ****local today, cloud tomorrow, back to local**** ? it's just a config/startup choice; the pure core gives ****equivalent**** output either way. ****Advertise it**** ("your machine *or* our cloud ? your call, switch anytime"). | owner 2026-06-21 | Honest caveat: equivalent at the ****rally/window**** level, not bit-identical across GPUs (D5); a single job runs in one profile, switching is ****between**** jobs. |

41 | D14 | ****NORTHSTAR ? operate toward this:**** a ****mobile, app-like**** flow ? point at a ****Google Drive**** video ? ****fully cloud-native**** run ? the stitched, ****ready-to-share**** reel lands ****back in Drive, next to the source****. | owner 2026-06-21 | Implies a ****`gdrive-api` (OAuth) StorageBackend**** (read source + write reel back) ? distinct from the local ****mount**** backend; ****resolves \$8 #7****, new ****M12****. The mobile UI is the khelsutra track; the engine must support it (async jobs + status + Drive round-trip / signed URL). Net-new scope (OAuth + Drive write); not necessarily v1, but the design must not preclude it (the ****`StorageBackend`** ABC accommodates it). |

42 | D15 | ****Auto-detect SUGGESTS, the user CONFIRMS ? execution is NEVER a surprise.**** A GPU owner may still choose cloud; ****cloud (= spend) is never entered silently****; the active profile is always explicit + surfaced. | owner 2026-06-21 | Amends \$4.3 (auto-detect = a proposed default, not an auto-decision). |

43

44 ## 3. Foundation ? evolution / ADR lineage ? *trace the idea end-to-end*

45 This project is the latest step in a long decision chain; each ADR contributed a piece and a ****subtlety that carries forward****:

46

47 | ADR / doc | Contributed | Subtlety carried into this project |

48 |---|---|---|

49 | ****ADR-005**** | Permissive licensing (MIT/Apache-2.0/BSD only) | The engine + any served model + the ****container image**** (ffmpeg, fonts, weights) stay commercial-clean. |

50 | ****ADR-008**** | Owned-model topology; ****train==serve featurizer single-sourcing**** | The "core unaffected across profiles" is the same parity discipline, extended from train/serve to ****local/cloud****. |

51 | ****ADR-009**** | KhelSutra Guru; ****local-first delivery**** | The ****truly-local profile**** is the direct continuation ? privacy/offline self-host stays first-class. |

52 | ****ADR-010**** | DECOUPLED: D0?GCP pivot | The cloud-compute origin (GPU + co-located bucket). |

53 | ****ADR-011**** | DECOUPLED scope = M0?M2+M3.5; ****deferred M3 (serving) + M4**

```

(billing/cost-guard)**; override the "rent nothing / not a service" posture | **This project
realizes M3/M4.** Carries $06's owner-gated gates: **DPA/consent for non-owner uploads,
collector `md5` keying, WASB-C3**. |
54 | **ADR-012** | GPU-native; TPU deferred; **Cloud Run+GPU scale-to-zero = serving
model**; **NVDEC** named as the decode lever | The serving substrate + the 7-step seed scope
this expands. |
55 | **ADR-013** | Drop D0; **GCP sole compute stack**; $200 D0 credits ? non-compute only
| The provider for cloud-serving; D0 Spaces remains only a possible S3 *storage* target. |
56 | **? ADR-014 (new)** | **Compute-decoupled serving:** one pure core, profile-selected
compute+storage; stitch-where-metered | This doc. **Refines** ADR-011/012, does **not**
supersede them. |
57 | PLATFORM_ARCHITECTURE.md | The `ComputeBackend`/`StorageBackend` registries | The
abstraction this realizes in code. |
58 | VIDEO_LOCALITY_MODEL.md / DESKTOP_APP_PLAN.md | L0 fully-local tier / `LocalDesktop`
backend | The truly-local profile = these, productionized. |
59 | 07-COMPUTE-ABSTRACTION (archived) | `DeviceContext`/`DeviceProfile` (designed) | WASB
hardcodes `device='cuda'` with **no CPU fallback** ? a portability gap M4 closes. |
60 | DEPLOY_REPORT_GCP_2026-06-20 (archived) | First real L4 run; **cross-GPU sub-pixel
parity finding** | Why D5 (parity at window level, not CSV bytes). |
61
62 ## 4. Design / architecture
63 See [architecture.svg](architecture.svg) for the picture. **The core never changes;
the box around it does.**
64
65 ### 4.1 The core contract (must stay byte/behaviour-identical) `[verified]`
66 1. **DETECT (GPU):** `DetectorRunner.run_predict(video_path, output_dir, video_id) ?
CSV[Frame,Visibility,X,Y]` (`backend/pipeline/detectors/base.py:164`). Three interchangeable
impls ? `WasbRunner` (WSL), `NativeWasbRunner` (Linux GPU), `StubDetectorRunner` (CPU mock) ?
emit the **identical CSV schema**. The trajectory CSV is the stable artifact boundary.
67 2. **WINDOW (CPU pure fn):** `trajectory_to_action_windows(points, fps, frame_width, ?)
? windows` (`tracknet_runner.py:281`). A shared `@staticmethod`, zero GPU/IO/randomness ? same
trajectory + config = byte-identical candidate windows. Used by every runner *and* the
eval/regression stack verbatim.
68 3. **STITCH (CPU/ffmpeg):** `VideoCompiler.compile_highlights(raw_video, segments,
out_name) ? reel` (`stitcher/compiler.py:303`). Pure FS+subprocess (merge ? ffmpeg ? watermark
? per-clip libx264 trim ? concat). Deterministic for a given input + ranges + watermark config.
69
70 **Non-goal (the trap to avoid):** *no* code branch inside detect/window/stitch keyed on
profile ? the same discipline the experiment harness already enforces (a disabled cue is byte-
identical to CONTROL).
71
72 ### 4.2 The deployment profiles
73 | Profile | Compute | Storage | Orchestration | When |
74 |---|---|---|---|---|
75 | **truly-local** | `compute_target=local_gpu`; `detector_impl=native` (Linux) / `auto`
(WSL); stitch **in-process** | `local` (or `gdrive` = mounted Drive) | `python -m backend.main`
or `worker infer` CLI; existing async job worker | Self-host / offline / privacy / dev-on-a-GPU-
box ? owner (b) |
76 | **cloud-serving** | `compute_target=cloud_run`; `detector_impl=native` (L4, cuda);
**detect AND stitch on the same Cloud Run job** | `gcs` (or gdrive mount / assembled upload);
per-video workspace; reels via signed URL | Cloud Run control plane enqueues ? Cloud Run GPU job
pulls/runs/pushes; `WAIT_AND_RESUME` reschedules the control-plane job; scale-to-zero | Hosted
SaaS ? owner (a) |
77 | **cpu-mock** | `compute_target=cpu_mock`; `detector_impl=stub` (synth/replay) |
`local`/in-memory fakes | `run_all_tests.py` / pytest | CI, regression, profile-parity, GPU-free
dev |
78 | *byo_worker* (deferred) | tenant GPU via outbound pull agent | tenant's own bucket
(signed URLs) | long-poll pull; same job table scoped by tenant | Privacy-sensitive enterprise ?
**DEFERRED**, enum/seam reserved |
79
80 ### 4.3 Profile selection (the "startup/setup config") `[design]`
81 ONE new top-level `deployment` section on `AppConfig` + preset application + env auto-
detect in `load_config`:
82 - `deployment.profile` {truly-local | cloud-serving | cpu-mock} (empty = today's
manual knobs, **fully backward-compatible**).

```

83 - Selecting a profile applies a **coherent preset bundle** of **existing** knobs that today drift independently: **INPUT** (new `input_backend`/input_root``, parallel to the output-side `StorageConfig``), **COMPUTE** (new `compute_target`` enum locking `detector_impl` + skip_ai_handoff` + workspace strategy`), **STORAGE** (`storage.backend`/workspace_root`/single_folder_per_video``, all already exist).

84 - **Precedence**: built-in defaults ? profile preset ? `config.json`` ? env (`RALLY_DETECTOR_IMPL``, `WASB_*`, `DEPLOYMENT_PROFILE``, ?) ? worker `--set k=v``. Every knob stays individually overridable.

85 - **Env auto-detect SUGGESTS**, the user **CONFIRMS** (D15 ? no surprise execution): the env gives a **proposed** default (`!K_SERVICE`` ? cloud-serving; `!CI=true`` ? cpu-mock; else `!torch.cuda.is_available()`` ? truly-local), but the user **explicitly confirms or overrides** it ? a GPU owner may still pick cloud, and **cloud (= spend) is NEVER entered silently**. The active profile is always surfaced (logged/shown). **Per-profile startup checks fail/warn fast** (cloud-serving without GCS creds/GPU ? loud error before any job).

86 - The worker (`cmd_infer`/cmd_train``) is **already profile-agnostic** (reads `config.storage``, accepts `--config/--set``) ? no worker rewrite.

87

88 **### 4.4 Compute placement ? and why stitch runs on Cloud Run** `[design]``

89 DETECT on GPU everywhere; WINDOW CPU-pure everywhere; STITCH is real CPU work. In **cloud-serving**, stitch is co-located in the Cloud Run job so `gpu_active_seconds`` (detect) + `wall_seconds`` (detect+stitch) + `bytes_out`` (reel egress) land on **one metered invocation** ? the per-job cost ledger. Running stitch on a laptop would leave that cost **invisible/unattributed** ? exactly what the owner wants to avoid. (Stitch stays synchronous in the GPU job for the common one-reel case; a queued CPU-only stitch fallback only if compiles get bursty ? the job table + `claim_due_job`` already supports both.)

90

91 **### 4.5 Reuse map ? most of the seams already exist** `[verified]``

92 **Exists**: the 3-stage core; `_resolve_runner`` (the single compute seam); `StubDetectorRunner`` (`replay_csv` = byte-exact $0 anchor`); `StorageBackend`` ABC + 4 backends + `StorageRef.parse_uri``; the worker CLI (`fetch/put, --config/--set``); the async job control plane (`jobs`` table, `claim_due_job`` CAS); `ExecutionPolicy WAIT_AND_RESUME`/JobWaiting`` (carries to Cloud Run dispatch verbatim); the chunked-upload router (`<2GB``); `skip_ai_handoff`` (LocalNoAI); storage fakes; golden fixtures + experiment harness. **Partial**: per-video workspace helper (default-OFF); `DeviceContext`` (designed); edge-auth (v4 `user_id`` columns exist). **Net-new**: the `deployment`` section; `input_backend``; configurable output base (kill hardcoded `output/``); Cloud Run dispatch shim + container image; cost ledger + pricebook; edge-auth header extraction; startup env detection.

93

94 **### 4.6 The data-flywheel harness** (D12 / M11) `[design]``

95 Gemini-on (D11) ships good reels now; this harness is **how we earn the right to flip Gemini off**. On a **consented, adult** cloud-serving job (the D10 trade), the pipeline additionally:

96 1. **Captures** the source upload + the Gemini reel + the rally output (intervals/report) into the per-video workspace (`workspaces/<video_id>/``), instead of the no-consent auto-delete path.

97 2. **Reliably stores** them durably in the bucket (the consented-retention bucket; raw video kept only for the training pool, not the no-consent path).

98 3. **Runs shadow evals** ? the owned model scores the same video alongside Gemini (`backend/eval/experiment.compare` + the dual-eval-set, [[eval-dual-set-regression]]`) ? tracks the owned-vs-Gemini gap per job (the live read on "is the floor reachable yet?").

99 4. **Promotes to golden** ? captured videos that pass review become **golden TRAINING rows** (human-reviewed labels via the annotation flow ? `data_pipeline/``), growing the corpus ? the owned model climbs to the D11 floor ? flip Gemini off via a Launch Report.

100 This is the consent?capture?eval?goldenize **flywheel** (the cloud realization of the PERSONALIZATION contribution model): every consented free reel makes the owned model better. Reuses `data_pipeline/``, `backend/eval/``, and the M9 consent gate; **adults-only**, raw-retention gated by the D10 consent.

101

102 **## 6. Honest limits ? what this does NOT do / prove**

103 - A green CI suite proves **plumbing + determinism** (window+stitch) + **profile-parity** for \$0 ? it does **NOT** prove field quality, nor that real-GPU detection is good (that's owner-side, real-video).

104 - **Cross-GPU detection is NOT byte-identical** ? parity is a window-level claim, not a CSV-byte claim.

105 - The cloud `0.1.0-l4`` weights are **n=2 plumbing-grade** ? cloud-serving ships with

****Gemini handoff ON**** until the owned model clears a floor (open Q).

106 - This does not resolve ****WASB-C3**** (sharing a trained model) or the ****DPA/consent**** gate ? both remain owner/counsel-gated before public, non-owner serving.

107

108 **## 7. Deliverables & progress tracker ? **source of truth****

109 Legend: ? Todo · ? In progress · ? Done · ? Gated. ****One small PR per row****, off `origin/master`, no stacking. Default-OFF where it touches core callers.

110

ID	Deliverable	Depends	Gated?	Status	PR
M0	This design doc + project dir + `runlogs/` + ADR-014			?	owner sign-off ?

****? APPROVED 2026-06-21**** |

114 | M1 | `deployment.profile` + `compute\_target` config + preset bundles + precedence/auto-detect; unit tests (no behaviour change, profile='' default) | M0 | safe | ? |

[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279) |

115 | M2 | `input\_backend`/`input\_root`; wire main CLI/API input through `StorageRef.parse\_uri` (local=identity); storage-fake tests | M1 | safe | ? |

[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280) |

116 | M3 | Configurable output/scratch base ? kill hardcoded `output/` (`trajectory\_hybrid:237,313`, `yolo\_hybrid:407`, `gemini`); ****DEFAULT stays `output/`****; golden + stub-E2E byte-identical | M1 | ? default-OFF + Launch Report on flip | ? |

[#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282) |

117 | M4 | `DeviceContext` + WASB ****CPU-fallback****; unified healthcheck/device wiring | M1 | safe (cuda default unchanged) | ? |

[#283](https://github.com/Khelsutra/badminton-highlight-indexer/pull/283) |

118 | M5 | ****truly-local profile**** end-to-end (preset + startup checks); first committed ****runlog**** (truly-local sample run) | M2,M3,M4 | owner real-GPU | ? | ? |

119 | M6 | Cloud Run GPU ****dispatch shim**** (control plane ? job via storage ? re-claim) + ****containerized worker image**** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner (GCP spend) | ? | ? |

120 | M7 | ****cloud-serving profile**** e2e on L4: upload`<2GB` + gdrive/bucket ? detect+stitch on Cloud Run ? reel via signed URL; ****DEPLOY_REPORT**** (posterity, sample runs + contact sheet) | M6 | ? owner (spend) | ? | ? |

121 | M8 | Per-job ****cost ledger**** (v_ migration: `owner\_id, gpu\_active\_seconds, wall\_seconds, bytes\_out, cost\_usd, cost\_breakdown\_json`) + ****pricebook**** + aggregation + `/api/.../cost` | M7 | ? owner (billing) | ? | ? |

122 | M9 | ****Edge auth**** (Cf-Access extraction) + per-tenant scoping on `user\_id`; DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? | ? |

123 | M10 | `byo\_worker` / zero-infra-BYO ? ****design-only, DEFERRED**** | M8,M9 | ? explicit owner approval | ? | ? |

124 | ****M11**** | ****Data-flywheel harness (D12):**** on a consented adult job, ****capture**** the upload + Gemini reel/rally output ? ****reliably store**** under the per-video workspace ? ****shadow-eval**** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? ****promote**** good ones to the golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the owned model climbs to the D11 floor. Reuses `data\_pipeline/` + `backend/eval/` + the consent gate (M9). | M7,M9 | ? owner (consent/privacy) | ? | ? |

125 | ****M12**** | ****gdrive-api` (OAuth) StorageBackend (D14 Northstar):\*\*** cloud reads the source from + writes the stitched reel **\*\*back to the user's Google Drive next to the source\*\*** (per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? | ? |

126

127 ****Testing strategy is a first-class deliverable**** ?

[`TESTING\_STRATEGY.md`](TESTING_STRATEGY.md). ****Run logs / sample runs for posterity**** ?

[`runlogs/`](runlogs/).

128

129 **## 8. Open questions ? owner *(blocking-gates vs nice-to-knows)***

130 ****? The 5 gating questions are LOCKED (owner, 2026-06-21) ? see §2 D7?D12:****

131 1. ? ****Cold-start ? D7**** (scale-to-zero, accept cold-start).

132 3. ? ****Pricebook ? D8**** (versioned DB table; USD internal / INR display).

133 4. ? ****Edge auth ? D9**** (Cloudflare Access; verify Cf-Access JWT, lock ingress to proxy).

134 5. ? ****DPA/consent ? D10**** (free-tier data-for-reels trade; adults-only; derived-data; counsel-gated to M9).

135 6. ? ****Quality floor ? D11**** + the ****data-flywheel harness D12 / M11**** (Gemini-ON until the owned model clears the floor; capture?store?eval?goldenize meanwhile).

```

136
137     **Still open (non-gating, smaller):**
138     2. **Stitch placement:** always co-located in the detect job (simplest, one metered
unit) vs split to a cheap CPU Cloud Run service when bursty? *Default co-located unless metrics
say otherwise.*
139     7. ? **RESOLVED ? D14:** cloud needs a **`gdrive-api` (OAuth) StorageBackend** (read the
source from + write the reel back to the user's Drive) ? the Northstar's Drive round-trip ?
distinct from the local **mount** backend (which stays for truly-local). New milestone **M12**.
140     8. **NVDEC/NVENC** in the Cloud Run image (cut stitch wall-time/cost) vs libx264
(determinism/parity)?
141     9. **Input cap:** is `<2GB` a hard client reject, or accept to the 4GB upload cap and
route larger files elsewhere?
142     10. **Directory name:** `COMPUTE_DECOUPLED_SERVING/` (chosen) vs keep
`CLOUD_RUN_GPU_SERVING/` ? confirm.
143
144     ## 9. Definition of done
145     - [ ] M1?M4 land (config + input + output-base + device layer), each green on golden
cross-OS + experiment no-regression.
146     - [ ] **Profile-parity test green** ? truly-local vs cloud-mock produce byte-identical
windows + segment ranges.
147     - [ ] **truly-local** (M5) and **cloud-serving** (M7) each have a committed **runlog /
DEPLOY_REPORT** with a real sample run.
148     - [ ] Cost ledger (M8) makes GPU + **stitch** + egress cost visible per job; edge auth
(M9) gates multi-tenant.
149     - [ ] **Data-flywheel harness (M11)** captures?stores?shadow-evals?goldenizes consented
adult jobs; the owned model is on a measured path to the D11 floor.
150     - [ ] ? flip Status DONE, archive to `docs/archives/past_projects/`.
151
152     ## 10. References
153     **ADR lineage:** ADR-005/008/009/010/011/012/013 ? **ADR-014**
(`docs/archives/decisions/DECISIONS.md`). **Strategy:** `PLATFORM_ARCHITECTURE.md`,
`HOSTING_PLAN.md`, `VIDEO_LOCALITY_MODEL.md`, `DESKTOP_APP_PLAN.md`,
`LOCAL_APPLIANCE_ARCHITECTURE.md` (khelsutra). **Archived predecessor:**
`docs/archives/past_projects/DECOUPLED_COMPUTE/` (esp. `05-COST-ATTRIBUTION.md`, `07-COMPUTE-
ABSTRACTION.md`, `06-RISKS-AND-OPEN-QUESTIONS.md`, `DEPLOY_REPORT_GCP_2026-06-20.md`). **Code
anchors `[verified]`:** `backend/pipeline/detectors/base.py:164`, `tracknet_runner.py:281`,
`stitcher/compiler.py:303`, `segmenters/trajectory_hybrid.py:65,237`, `backend/storage/*`,
`backend/worker/__main__.py:88`, `backend/api/{uploads,job_worker}.py`,
`backend/infrastructure/database.py`. **Design provenance:** workflow `wf_56274ff0` (4 code-
mappers ? synthesis; 58 findings).
154
155     ---
156     ### Changelog
157     - 2026-06-21 ? created from the design workflow output; M0 in progress.
158     - 2026-06-21 ? **M0 APPROVED**; **M1 landed**
([#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279)):
`deployment.profile` + `compute_target` + preset bundles + precedence/auto-detect-as-suggestion
(D15) on `AppConfig`. Pure/additive, `profile=''` default == today's behaviour (zero core
change); presets touch only existing knobs (no M2 `input_backend` / M3 output-base over-reach).
Suite 1068 green, cov 84.11%.
159     - 2026-06-21 ? **M2 landed** ([#280](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/280)): `input_backend`/`input_root` + URI-aware main API/CLI input via
`resolve_input_uri`/`acquire_local_input` (local = identity passthrough, byte-for-byte
unchanged; bucket/Drive fetched to scratch). Unsupported scheme ? clean 400; `local://` accepted
on the resolved key; hosted-mode rejects non-local URIs; source URI kept for DB/report
provenance. Suite 1105 green, cov 84.84%.
160     - 2026-06-21 ? **M3 landed** ([#282](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/282)): configurable output/scratch base via
`backend/utils/paths.output_base(config)` ? the 3 segmenters' chunk/handoff/temp dirs root under
`output.output_dir` (default `"output"`, byte-identical; default-OFF ? no Launch Report). Fixes
the latent bug where scratch ignored `--output-dir`. Suite 1114 green, cov ~84.9%.
161     - 2026-06-21 ? **M4 landed** ([#283](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/283)): `DeviceContext` + native WASB **CPU-fallback** via a new `device="auto"`
(cuda-if-available-else-cpu); explicit `"cuda"` stays strict (no silent slow-CPU downgrade),
default unchanged. `"auto"` resolves before argv (never reaches `wasb_infer.py`); CUDA probe

```

cached. Device-selection tested offline; real CPU-inference run owner/GPU-verified. Suite 1126 green, cov 84.96%. **? M1?M4 (the config/input/output/device layer) COMPLETE.**
162

10. user (2026-06-21T09:25:12.442Z)

```
backend\pipeline\segmenters\gemini.py:168:         chunks_dir: str,
backend\pipeline\segmenters\gemini.py:186:         os.makedirs(chunks_dir, exist_ok=True)
backend\pipeline\segmenters\gemini.py:192:         chunk_path = os.path.join(chunks_dir,
f"chunk_{video_id}_{ci}.mp4")
backend\pipeline\segmenters\gemini.py:314:         os.rmdir(chunks_dir)
backend\pipeline\segmenters\gemini.py:540:         chunks_dir = _chunk_workdir(video_id,
output_base(self.config))
backend\pipeline\segmenters\gemini.py:545:         chunk_duration, chunk_overlap,
ai_scale_width, chunks_dir, cleanup_temp_files,
backend\pipeline\segmenters\trajectory_hybrid.py:314:         handoff_dir =
os.path.join(output_base(self.config), f"chunks_{provider_name}_handoff")
backend\pipeline\segmenters\yolo_hybrid.py:102:         frame_skip: int, chunks_dir: str,
backend\pipeline\segmenters\yolo_hybrid.py:133:         chunk_path =
os.path.join(chunks_dir, f"chunk_{video_id}_{ci}.mp4")
backend\pipeline\segmenters\yolo_hybrid.py:171:         chunk_path =
os.path.join(chunks_dir, f"chunk_{video_id}_{ci}.mp4")
backend\pipeline\segmenters\yolo_hybrid.py:269:         gemini_dir =
os.path.join(output_base(self.config), f"chunks_{provider_name}_handoff")
backend\pipeline\segmenters\yolo_hybrid.py:408:         chunks_dir =
os.path.join(output_base(self.config), "chunks_yolo")
backend\pipeline\segmenters\yolo_hybrid.py:409:         os.makedirs(chunks_dir, exist_ok=True)
backend\pipeline\segmenters\yolo_hybrid.py:415:         frame_skip, chunks_dir,
backend\pipeline\segmenters\yolo_hybrid.py:424:         os.rmdir(chunks_dir)
```

11. assistant (2026-06-21T09:25:17.107Z)

Key finding confirmed: M3 (#282) introduced `backend/utils/paths.output\_base(config)` and the `output/chunks\_\*` dirs now root under `output\_base()`, but they are still **hardcoded folder names** (`chunks\_{provider}\_handoff`, `chunks\_yolo`). Let me verify the `paths.py` helper and check the P3 reconcile precisely, plus grep the remaining cross-track items (WASB-C3, collector md5, DO droplet/token, repo rename, Ops-Agent).

12. user (2026-06-21T09:25:17.786Z)

```
1      """Path-safety helpers for the API boundary (untrusted request fields)."""
2      import os
3      from typing import Any, Optional
4
5
6      def output_base(config: Any) -> str:
7          """The configured base dir for a segmenter's LOCAL scratch/staging artifacts
8              (COMPUTE_DECOUPLED_SERVING M3). Returns ``output.output_dir`` (default ``"output"``)
so the
9          chunk-staging / AI-handoff / temp dirs follow ``--output-dir`` /
``output.output_dir`` instead of
10         a hardcoded ``./output``. The DEFAULT is ``"output"`` ? byte-identical to the prior
literal.
11
12         This is the LOCAL output dir, deliberately NOT the storage ``workspace_root`` (which
may be a
13         ``gs://`` URI you can't ``makedirs``). Accepts the typed AppConfig or the legacy
dict-shim
14         (``.get("output", {}), {}).get("output_dir")``); any miss or empty value falls back to
``"output"``. """
15         try:
16             out = config.get("output", {}).get("output_dir")
17         except Exception:
18             out = None
```

```

19         return out or "output"
20
21
22     def safe_output_filename(name: str) -> str:
23         """Validate an output filename is a BARE name (no dir part, no traversal, not
absolute).
24
25         `os.path.join(output_dir, name)` silently discards `output_dir` when `name` is
absolute,
26         and `..` segments escape it ? so an unchecked `output_filename` is an arbitrary-
write
27         primitive. Returns the name unchanged when safe; raises ValueError otherwise.
28         """
29         if not name or not name.strip():
30             raise ValueError("output_filename must not be empty")
31         if "\x00" in name:
32             raise ValueError("output_filename must not contain a null byte")
33         if "/" in name or "\\" in name:
34             raise ValueError(f"output_filename must not contain a path separator: {name!r}")
35         if os.path.isabs(name) or name != os.path.basename(name) or name in (".", ".."):
36             raise ValueError(f"output_filename must be a bare filename, got: {name!r}")
37         return name
38
39
40     def resolve_under_root(path: str, root: str) -> str:
41         """Return the real path, asserting it is contained under `root` (raises ValueError
if not)."""
42         rp = os.path.realpath(path)
43         rr = os.path.realpath(root)
44         try:
45             contained = os.path.commonpath([rp, rr]) == rr
46         except ValueError: # different drives on Windows
47             contained = False
48         if not contained:
49             raise ValueError(f"path {path!r} escapes the allowed root {root!r}")
50         return rp
51
52
53     def validate_input_path(path: str, allowed_root: Optional[str] = None) -> None:
54         """Reject null bytes and (when an allowed_root is configured) out-of-root inputs.
55
56         With no allowed_root (default, single-user local mode) only the null-byte check
applies,
57         so the local 'point it at any of my videos' workflow is preserved. Setting an
58         allowed_root (a future hosted/tenant deployment) enforces containment.
59         """
60         if "\x00" in path:
61             raise ValueError("video_path must not contain a null byte")
62         if allowed_root:
63             resolve_under_root(path, allowed_root)
64

```

13. user (2026-06-21T09:25:19.182Z)

docs\DOC_STATUS.md:26:> **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** **?**
verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl.
carried-forward WASB-C3/DPA/\`md5`), reindexed `docs/README` + `past\_projects/README`, kept the
ADR-010?014 lineage, then `git mv` to `docs/archives/past\_projects/DECOUPLED\_COMPUTE/`.

docs\DOC_STATUS.md:56:| `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |
docs\COMMERCIALIZATION.md:42:[Omitted long matching line]
docs\COMMERCIALIZATION.md:65:| **WASB-SBDT** (shuttle) | MIT (code) | No | ? code clear |
Weights provenance open (C3). |
docs\COMMERCIALIZATION.md:74:and **weights provenance (C3)**, plus the metered-API privacy
posture (C4).
docs\COMMERCIALIZATION.md:193: confirm each (C3).

docs\NEXT_STEPS.md:26:13. ****WASB-C3**** weight provenance (gates ***sharing*** a trained model) ? legal.

docs\NEXT_STEPS.md:133: by C3 (provenance multiplies per sport).

docs\ANALYZERS_AND_RECONCILERS.md:639:- ****C3 ? out-of-fold (honest) stacking.**** When a producer is itself learned (shuttle-state), feed the LR its

docs\ANALYZERS_AND_RECONCILERS.md:687:- ****Stacking leakage**** (in-sample learned-producer outputs) ? C3; and ****threshold-tuning-on-test**** ? keep

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:77: is per-sport detectors: **** WASB (badminton) has the unresolved ****C3**** checkpoint-provenance issue, and ****clean MIT/****

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:121:- **** (ii) WASB C3 ? first-class action item (owner): **** the academic-data checkpoint provenance is a ****commercial-ship******

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:156:| 2 | WASB C3 ? first-class action item (own-weights / clean swap) | ****owner: yes**** ? |

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:32:- ****HIGH****: anything touching "paid-LLM as training data" guardrail, unquantified large costs (Gen-2 VRAM/\$), production default flips without lift proof, licensing/C3 blockers, multisport generalization hypothesis.

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:64:6c. ****WASB C3 + related commercial blockers (tracking only)**** ? first-class tracked item (own-weights or clean detector; multiplies per sport). No code change in this repo until resolved upstream.

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:68: Ingest Roora pickleball/TT CSVs (`import-local --normalize`). Merged-prototype LOVO across badminton+TT once TT has ?3 matches. Identify clean MIT/Apache/BSD ball/shuttle detector per sport (WASB C3 is unresolved blocker for some paths; TT-on-WASB serving gated). Single-model-multisport-temporal-generalization is unproven hypothesis (validate on 2nd-sport corpus).

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:69: ****Risk: MEDIUM**** (data + per-sport detector hunt; C3 is commercial blocker).

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:98:| 6c | C3 / commercial blockers | (tracking) | Multiplies per sport; gates ship | Tracked in BACKLOG/OWNED; no code until resolved |

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:99:| 7 | Multisport + detectors | MEDIUM | Data + per-sport detector hunt (C3 multiplies) | ?3 TT matches + cross-LOVO prototype |

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:135:- 6a/6c. Gemini-silver purge (moat code) + C3 tracking / commercial resolution.

docs\PROMINENT_COURT_DETECTION.md:105:| C3 | GX010142 litmus: rally #1 dropped, central rallies kept (measured) | C2 | No | ? | ? |

docs\PROMINENT_COURT_DETECTION.md:109:| C7 | Launch decision (default-ON) with a Launch Report | C3,C5 | ? owner | ? | ? |

docs\COMPUTE_DECOUPLED_SERVING\README.md:53:| ****ADR-011**** | DECOUPLED scope = M0?M2+M3.5; ****deferred M3 (serving) + M4 (billing/cost-guard)****; overrode the ****rent nothing / not a service**** posture | ****This project realizes M3/M4.**** Carries \$06's owner-gated gates: ****DPA/consent for non-owner uploads, collector `md5` keying, WASB-C3****. |

docs\COMPUTE_DECOUPLED_SERVING\README.md:106:- This does not resolve ****WASB-C3**** (sharing a trained model) or the ****DPA/consent**** gate ? both remain owner/counsel-gated before public, non-owner serving.

docs\COMPUTE_DECOUPLED_SERVING\TESTING_STRATEGY.md:61:- gotchas (conda-ToS, WASB-C3, GPU stockouts, font fallback) + the ****final teardown state**** (\$0 audit);

docs\QUALITY_ITERATIONS.md:645:- ****C3 ? honest stacking.****

`eval\_stats.out\_of\_fold\_predictions`/'grouped_folds` (leave-one-group-out by

docs\RALLY_DETECTION_QUALITY_REPORT.md:366:- ****C3 ? WASB checkpoint provenance**** is a commercial-ship blocker a clean head does not launder; an

docs\COMPUTE_DECOUPLED_SERVING\runlogs\README.md:24:6. ****Gotchas**** ? conda-ToS, WASB-C3, GPU stockouts, font fallback, cold-start, NVDEC, etc.

docs\archives\decisions\DECISIONS.md:418: WASB C3 checkpoint status). The loader/gate hard-fails on schema mismatch ? the

docs\archives\decisions\DECISIONS.md:422: The gate refuses commercial ship while C3 is unresolved, regardless of F1.

[Showing results with pagination = limit: 30]

14. assistant (2026-06-21T09:25:22.462Z)

Now let me grep the remaining cross-track items: collector md5 export, DO droplet/token retirement, repo rename off "badminton", and the Ops-Agent codification into deploy/gcp.

15. user (2026-06-21T09:25:24.499Z)

```
.gitattributes:7:# Shell scripts must stay LF so they run on Linux boxes (the DigitalOcean
.gitattributes:9:# (`#!/usr/bin/env bash\r`) fails on Linux. See deploy/digitalocean/.
CLAUDE.md:59: Cloud Run+GPU for serving per ADR-012). **DigitalOcean is DROPPED for compute**
(DO GPU not enabled
CLAUDE.md:62: `deploy/digitalocean/` scripts are kept solely as provider-agnostic Linux-GPU
setup reused by GCP.
artifacts\appliance_design.json:9:[Omitted long matching line]
artifacts\so.py:33: "Operationally proven: the exact box (DO droplet) already co-hosts
site+engine and ran the full train+infer e2e against real Spaces with the GPU mocked - this
design is the UI on top of a verified substrate."
artifacts\so.json:1:[Omitted long matching line]
backend\worker\__main__.py:11:the path validated end-to-end on DigitalOcean
(`docs/SETUP_NEW_MACHINE.md`).
deploy\gcp\README.md:3:> **? GCP is the SOLE compute stack (ADR-013, 2026-06-20).** DigitalOcean
is **dropped for compute**
deploy\gcp\README.md:6:> The `deploy/digitalocean/` scripts are kept only as **provider-
agnostic** Linux-GPU setup that this
deploy\gcp\README.md:11:(non-mock) WASB training + inference. This is the **DO?GCP pivot**
(ADR-010): DigitalOcean GPU
deploy\gcp\README.md:77:> BOTH training and serving run on GCP; DigitalOcean is dropped for
compute.** Spot is still cheaper
deploy\gcp\README.md:148:The `deploy/digitalocean/` scripts are **provider-agnostic** (they
detect the local NVMe / install
deploy\gcp\README.md:165:sudo bash deploy/digitalocean/mount_scratch.sh && export
TMPDIR=/scratch # detects the local NVMe
deploy\gcp\README.md:166:./deploy/digitalocean/bootstrap.sh --gpu
# venv + torch cu124 (confirm cuda True)
deploy\gcp\README.md:168:bash deploy/digitalocean/gpu_setup.sh
# native `wasb` conda env + WASB-SBDT
deploy\digitalocean\README.md:1:# Testing the rally pipeline on DigitalOcean
deploy\digitalocean\README.md:3:> **? DEPRECATED (ADR-013, 2026-06-20) ? DigitalOcean is DROPPED
for compute.** Use
deploy\digitalocean\README.md:11:A storage-free, secret-free way to get the pipeline running on
a DigitalOcean (or Paperspace)
deploy\digitalocean\README.md:25:| 2 | **DigitalOcean API token** | DO console ? API ? Tokens ?
Generate (Personal Access Token) | **read + write** (Droplets/Spaces); scope down if the UI
allows | **now** (for `doctl`; optional if you click-create the box) |
deploy\digitalocean\README.md:31:**The "API key that drives the GPU"** in the North-Star sense
is **2 (DO token)** + **3 (Paperspace key)** ?
deploy\digitalocean\README.md:56:./deploy/digitalocean/bootstrap.sh # CPU box
deploy\digitalocean\README.md:58:./deploy/digitalocean/bootstrap.sh --gpu # GPU box
(installs CUDA torch)
backend\storage\s3.py:3:One backend covers **AWS S3, Cloudflare R2, DigitalOcean Spaces, and
MinIO** ? they differ
deploy\digitalocean\bootstrap.sh:3:# Bootstrap a DigitalOcean / Paperspace Ubuntu box to run the
rally pipeline.
deploy\digitalocean\bootstrap.sh:5:# ./deploy/digitalocean/bootstrap.sh # CPU box
(Droplet) ? M0 / M1-cpu
deploy\digitalocean\bootstrap.sh:6:# ./deploy/digitalocean/bootstrap.sh --gpu # GPU box
(Paperspace / GPU Droplet) ? M1 / M3.5
docs\archives\decisions\DECISIONS.md:511:## ADR-010: DECOUPLED_COMPUTE ? pivot GPU + storage
from DigitalOcean to GCP (asia-south1)
docs\archives\decisions\DECISIONS.md:517:1. The DECOUPLED_COMPUTE **GPU box + co-located object
storage move from DigitalOcean to GCP**,
docs\archives\decisions\DECISIONS.md:528:- **DigitalOcean GPU Droplets are North-America-only**
(`nc2`/`tor1`/`atl1`) ? DO has **no
docs\archives\decisions\DECISIONS.md:553:**Date:** 2026-06-20 . **Status:** RATIFIED (owner, in-
session). **Refines ADR-010.** ? **The DO=training / GCP=serving split is SUPERSEDED by
[ADR-013](#adr-013-drop-digitalocean-for-compute--gcp-is-the-sole-compute-stack-training--
serving)** ? DigitalOcean is dropped for compute; GCP is the sole compute stack. (The scope
decision ? DECOUPLED = M0?M2 + M3.5 ? still stands.)
docs\archives\decisions\DECISIONS.md:566: - **DigitalOcean = TRAINING / the decoupling core.**
Use the **$200 DO credits** ; DO GPU droplets
docs\archives\decisions\DECISIONS.md:661:## ADR-013: Drop DigitalOcean for compute ? GCP is the
SOLE compute stack (training + serving)
```

docs\archives\decisions\DECISIONS.md:667:1. ****DigitalOcean is DROPPED as a compute provider.****
****GCP is the sole compute stack**** for BOTH training and serving. (ADR-011's "DO=training,
GCP=serving" split is retired ? GCP does both. The DECOUPLED scope decision in ADR-011 ? M0?M2 +
M3.5 ? still stands.)

docs\archives\decisions\DECISIONS.md:674:- ****Paperspace (now DO Gradient) free GPU is
subscription-gated**** (a paid "G"/Growth plan; high-end A100/H100 needs the \$39/mo Growth plan),
and Paperspace bills ****separately**** from DO Droplet credits ? the \$200 credit doesn't fund it
either.

docs\archives\decisions\DECISIONS.md:679:- ****Compute provisioning = `deploy/gcp/` only.****
`SETUP\_NEW\_MACHINE.md` repointed to GCP; `deploy/digitalocean/README.md` marked ****DEPRECATED****;
`HOSTING\_PLAN.md`'s "All-DigitalOcean" compute option retired.

docs\archives\decisions\DECISIONS.md:680:- ****`deploy/digitalocean/` scripts are KEPT**** ?
`bootstrap.sh` / `mount\_scratch.sh` / `gpu\_setup.sh` are ****provider-agnostic**** Linux-GPU setup
the GCP runbook reuses; only the DO-specific ***provisioning*** (doctl droplet create, Spaces) is
dropped. (Optional later cleanup: move them to `deploy/common/`.)

docs\archives\decisions\DECISIONS.md:681:- The shared DO CPU droplet ****`claude-do-rally-test`****
(another session's testbed) is ****owner-coordinated**** cleanup, separate from this decision; the
freshly-rotated ****DO API token**** can be retired once that droplet is gone.

docs\archives\past_projects\README.md:12:- **`DECOUPLED\_COMPUTE/`** ? ? ****DELIVERED + ARCHIVED
2026-06-21.**** Lifted the rally pipeline onto a rented cloud GPU (bucket-driven train + infer);
****M0?M2 + M3.5 proven on a GCP L4**** (M2 ? `weights/0.1.0-l4/`). Delivered on ****GCP**** ?
DigitalOcean dropped for compute (ADR-013). ****M3/M4 (serving endpoint + per-video billing +
scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject**** (ADR-012). The full deploy
record is **`DECOUPLED\_COMPUTE/DEPLOY\_REPORT\_GCP\_2026-06-20.md`**.

docs\HOSTING_PLAN.md:4:****Date:**** 2026-06-10 · ****Constraints:**** budget \$10?25/mo · GCP credits
available · DigitalOcean

[Showing results with pagination = limit: 40]

16. user (2026-06-21T09:25:25.899Z)

docs\BACKLOG.md:13:- ? ****Rename the repo off "badminton"**** ? owner-emphatic "ASAP" action; the
project is multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs.
****Revisit:**** owner action, before broad external sharing.

docs\archives\decisions\DECISIONS.md:424: **`sports-temporal-models`**, to be finalized with the
pending repo rename) ****when****:

docs\NEXT_STEPS.md:27:14. ****Rename repo off "badminton"**** (owner, BACKLOG).

docs\NEXT_STEPS.md:62:****Leaving (bulky / M2 ? no start without owner go):**** ActionFormer (M0.4),
event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.

docs\NEXT_STEPS.md:145: deletion in AI Studio); ****rename repo off "badminton"**** (owner action,
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:159:| 5 | Rename repo off "badminton" | ****owner action****
? |

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:71:8. ****Housekeeping ? rename repo
off "badminton" (BACKLOG ? + OWNED)****

docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:126:***Parallel / owner-led (not
primary agent engineering focus):*** Grow the golden corpus (your #1 leverage item), PMF field
visits (using the existing kit), repo rename. These run alongside M0 without blocking the quick
engineering wins.

17. user (2026-06-21T09:25:27.545Z)

deploy\gcp\README.md:113:### 3b. Observability ? Ops Agent (GPU graphs + logs in the console)

deploy\gcp\README.md:114:The VM ****Observability ? GPU**** graphs + Logs need the ****Ops Agent****
(the console's OS-policy install is

deploy\gcp\README.md:126:curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-
repo.sh && sudo bash add-google-cloud-ops-agent-repo.sh --also-install

deploy\gcp\README.md:127:# (if the agent was installed BEFORE the APIs were enabled: `sudo
systemctl restart google-cloud-ops-agent`)

deploy\gcp\README.md:129:> Best: bake `curl ?add-google-cloud-ops-agent-repo.sh|bash --also-
install` into the VM

18. assistant (2026-06-21T09:25:30.011Z)

Let me look at the deploy/gcp scripts to confirm the Ops-Agent codify status (is it in a startup-script, or only documented in README?), and check the collector md5 export reference.

19. user (2026-06-21T09:25:30.759Z)

```
deploy\gcp\provision.sh
deploy\gcp\teardown.sh
deploy\gcp\README.md
```

20. user (2026-06-21T09:25:32.378Z)

```
docs\BACKLOG.md:49:[Omitted long matching line]
docs\BACKLOG.md:98: write-only `_runtime_overrides` hack. Same pass also unified `video_id` MD5
derivation into
docs\CODE_MAP.md:55: -> video_id = MD5(whole file)
@backend/utils/hashing.py::compute_video_id (single shared helper, 64KB chunks; imported by
main + all run_*.py ? no more per-entrypoint copies)
docs\CODE_MAP.md:115:[Omitted long matching line]
docs\CODE_MAP.md:120:- `@run_eval.py::main` ? score one video's DB preds vs GT CSV.
`--stage{candidates,final,both}`. Exit 2=arg/IO. ? pure scorer; errors if video never processed.
`::default_db_path` resolves via `backend.config.load_config`
(`output.output_dir`/`output.db_name`). `get_file_md5` is an alias for `compute_video_id`.
docs\CODE_MAP.md:121:- `@scripts/run_phase3_eval.py::main` ? walk manifest, segment-if-needed,
score, aggregate. `load_dotenv()` at import. ? DB key = file MD5, NOT manifest's `video_id`
(kept as label); config via `backend.config.load_config`; `segmenter_cls(db, cfg)` receives the
typed `AppConfig` (same object the live pipeline feeds segmenters).
docs\CODE_MAP.md:183:[Omitted long matching line]
docs\CODE_MAP.md:252:$ **collector manifest.json** (eval): `{eval_sets:[{video_id(label),
local_path, csv}, ...]}`; ? DB key = file MD5, manifest `video_id` is a label.
docs\CODE_MAP.md:275:- Indexer keys video by **FILE MD5**; collector keys by human `video_id`. ?
re-encoding/re-downloading changes bytes -> MD5; acquisition MUST precede processing; manifest
tracks the exact file used.
docs\CODE_MAP.md:277:- Collector also emits golden rally labels
(`rally_number,start_time,end_time,...`) -> consumed by `@backend/eval/calibrate_wasb.py` /
`export_wasb_segments.py` / `@backend/tools/rally_slicer.py`.
docs\CODE_MAP.md:282:- **MD5 keying:** `video_id` = MD5 of WHOLE file via the single
`@backend/utils/hashing.py::compute_video_id` helper (64KB chunks), imported by main + all
run_*.py. Manifest `video_id` is a label, NOT the DB key.
docs\CODE_MAP.md:358:| `test_eval_harness.py` | `eval.harness` DB-pred fetch + evaluate +
report_to_dict; also `@run_eval.py::main`/`get_file_md5` (exit codes, `--scaffold`/`--json`) |
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:77:- 2026-06-10 hardening
(detector keying by video_id/MD5, re-ingest destroy-AFTER-confirm via watermarks+prune,
WAL+busy_timeout, typed wasb config, ExecutionPolicy, path safety helpers, single
`compute_video_id`, trajectory twin dedup into `TrajectoryHybridSegmenter`, person cue
extraction to reusable producer) is visibly landed and tested.
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:229:python run_regression.py
--video-id <known-good-md5> # must exit 0
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md:194:python run_regression.py
--video-id <known-good-md5> # must exit 0
docs\archives\PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md:45: sanitized stem +
`video_id[:12]` (the MD5 already used as the cache key), e.g.
docs\archives\decisions\DECISIONS.md:81: by **file MD5**. A re-encode/re-download changes the
bytes, hence the MD5.
docs\archives\decisions\DECISIONS.md:92:3. Honour the **MD5 keying** contract:
evaluation/processing must run on the exact video
docs\archives\decisions\DECISIONS.md:427: AND (c) the harness's only data input is the
collector's M0.1 export. By then the artifact
docs\archives\decisions\DECISIONS.md:433:round-trip test against the collector's `curate export`
format. Every ship-gate quantity
docs\archives\decisions\DECISIONS.md:691:[Omitted long matching line]
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md:68:- Add a `split: "train" |
"eval"` field to the collector's `curate export`
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md:72: file MD5 ? re-encoded
duplicates share a match but differ in
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md:82:- MD5-only dedup is
```


insufficient (re-encodes evade it) ? must track match identity.

docs\GOLDEN_REGRESSION_FIXTURES.md:96:- ****Opaque RANDOM surrogate id**** (`os.urandom` + a private surrogate?source map kept off-git) ? ***not*** the MD5 `video\_id`, and ***not*** HMAC(salt, video_id?name) (red-team: brute-forceable over a tiny known roster if the salt leaks). Never persist the source `name`/filename.

[Showing results with pagination = limit: 25]

21. assistant (2026-06-21T09:25:42.212Z)

The collector md5 export is a sibling-repo (sports-data-collector) action, not in this repo. Let me check the deploy/gcp/provision.sh to confirm whether the Ops-Agent is already baked into the startup script or still only documented in the README (this determines if the codify item is real Claude-doable work).

22. user (2026-06-21T09:25:43.015Z)

```
1      #!/usr/bin/env bash
2      # Provision the DECOUPLED_COMPUTE GCP storage + IAM (asia-south1 / Mumbai). Idempotent:
safe to
3      # re-run; already-existing resources are left as-is. The GPU VM is NOT created here (it
is gated on
4      # the GPUS_ALL_REGIONS quota ? see deploy/gcp/README.md §2-3); create it with the §3
command once
5      # the quota is granted. This documents EXACTLY what was run on 2026-06-20 (D0->GCP
pivot, ADR-010).
6      #
7      # Prereq: `gcloud auth login`; Service Usage API + a linked billing account enabled in
the console
8      # (README §1). Run: bash deploy/gcp/provision.sh
9      set -uo pipefail
10
11     PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
12     REGION="${RALLY_GCP_REGION:-asia-south1}"
13     BUCKET="${RALLY_GCS_BUCKET:-khehsutra-rally-corpus}"
14     BILLING_ACCT="${RALLY_BILLING_ACCT:-01420D-419EE0-53D83C}"
15     SA_NAME="rally-worker"
16     SA_EMAIL="${SA_NAME}@${PROJECT}.iam.gserviceaccount.com"
17     SA_KEY="${RALLY_SA_KEY:-$HOME/.gcp/rally-worker-sa.json}"
18     export CLOUDSDK_CORE_DISABLE_PROMPTS=1
19
20     echo "== project=$PROJECT region=$REGION bucket=gs://$BUCKET =="
21
22     echo "== enable APIs =="
23     gcloud services enable compute.googleapis.com cloudbilling.googleapis.com \
24         billingbudgets.googleapis.com iam.googleapis.com cloudresourcemanager.googleapis.com \
25         --project "$PROJECT"
26
27     echo "== GCS bucket (asia-south1, uniform bucket-level access) =="
28     gcloud storage buckets create "gs://$BUCKET" --location="$REGION" \
29         --uniform-bucket-level-access --project "$PROJECT" || echo " (bucket exists ? ok)"
30
31     echo "== service account + Storage Object Admin + key =="
32     gcloud iam service-accounts create "$SA_NAME" --display-name="Rally worker (storage)" \
33         --project "$PROJECT" || echo " (SA exists ? ok)"
34     gcloud projects add-iam-policy-binding "$PROJECT" \
35         --member="serviceAccount:$SA_EMAIL" --role="roles/storage.objectAdmin"
--condition=None \
36         --format="value(etag)" >/dev/null
37     if [ ! -f "$SA_KEY" ]; then
38         mkdir -p "$(dirname "$SA_KEY")"
39         gcloud iam service-accounts keys create "$SA_KEY" --iam-account="$SA_EMAIL"
40         echo " key -> $SA_KEY (NEVER commit this; export
GOOGLE_APPLICATION_CREDENTIALS=$SA_KEY)"
41     else
```

```

42     echo "   key already at $SA_KEY ? leaving it (delete to rotate)"
43 fi
44
45     echo "== \$100 budget alert (50/90/100%) =="
46     gcloud billing budgets create --billing-account="$BILLING_ACCT" \
47         --display-name="rally-100usd-guard" --budget-amount=100 \
48         --filter-projects="projects/$PROJECT" \
49         --threshold-rule=percent=0.5 --threshold-rule=percent=0.9 --threshold-rule=percent=1.0
50     || echo " (budget may already exist ? check: gcloud billing budgets list --billing-
account=$BILLING_ACCT)"
51
52     echo "== GPU quota (the VM gate) =="
53     echo "   regional $REGION:"
54     gcloud compute regions describe "$REGION" --project "$PROJECT" \
55         --flatten="quotas[]" --format="table(quotas.metric,quotas.limit)" | grep -E
"NVIDIA_(T4|L4)_GPUS " || true
56     echo "   global:"
57     gcloud compute project-info describe --project "$PROJECT" \
58         --flatten="quotas[]" --format="table(quotas.metric,quotas.limit)" | grep
GPUS_ALL_REGIONS || true
59     echo ""
60     echo "If GPUS_ALL_REGIONS == 0, request an increase (README §2), then create the VM
(README §3)."
```

23. assistant (2026-06-21T09:25:49.124Z)

Confirmed: `provision.sh` provisions only storage + IAM + budget; the GPU VM is created via a manual README §3 `gcloud` command and the Ops-Agent is documented only in README §3b (lines 113-129), **not** codified into a startup-script or the provisioning. So the Ops-Agent codify item is real, Claude-doable work (write a `startup-script` snippet / VM-create wrapper in `deploy/gcp/` that bakes the 3 pieces). Let me check the README §3 VM-create command and whether a startup-script hook exists, plus confirm whether `provision.sh` enables the logging/monitoring APIs (one of the "3 pieces").

24. user (2026-06-21T09:25:49.930Z)

```

60     ``
61
62     **Owner:** IAM & Admin ? **Quotas** ? filter **"GPUS (all regions)"** ? Edit ? request
**1** ? submit.
63     (For a single T4 this is usually approved in minutes; sometimes a short review.) Re-
check:
64
65     ``bash
66     gcloud compute project-info describe --project gen-lang-client-0306026826 \
67         --flatten="quotas[]" --format="table(quotas.metric,quotas.limit)" | grep
GPUS_ALL_REGIONS
68     ``
69
70     When that reads `1`, run §3.
71
72     > **? Spot does NOT skip the quota.** TESTED 2026-06-20: a **Spot** T4 create also fails
with
73     > `Quota 'GPUS_ALL_REGIONS' exceeded. Limit: 0.0 globally` ? `GPUS_ALL_REGIONS` gates
**both**
74     > on-demand AND preemptible GPUs, so the increase above is required either way. (A
pending request
75     > was filed via the Cloud Quotas API: `POST ?/quotaPreferences` `GPUS-ALL-REGIONS-per-
project`
76     > ? 1.) **Note (2026-06-20, ADR-013 supersedes the ADR-011 split): GCP is the SOLE
compute stack ?
77     > BOTH training and serving run on GCP; DigitalOcean is dropped for compute.** Spot is
still cheaper
```

```

78     > once the quota is granted, and the resumable cache makes preemption recoverable.
79
80     ---
81
82     ## 3. Create the GPU VM (after quota ? 1)
83
84     T4 is plenty for WASB + Gen-0. A local SSD gives a fast `/scratch`. *(The native runner
defaults to
85     **streaming** ? it decodes frames in-process, skipping the slow cv2 PNG extraction; bit-
identical to disk,
86     verified on a real GPU 2026-06-20. Set `indexing.wasb.stream_video=false` only if a
container/codecs cv2
87     can't seek; `WASB_SCRATCH` still backs the disk path.)*
88
89     ```bash
90     gcloud compute instances create rally-gpu \
91         --project=gen-lang-client-0306026826 \
92         --zone=asia-south1-a \
93         --machine-type=n1-standard-8 \
94         --accelerator=type=nvidia-tesla-t4,count=1 \
95         --maintenance-policy=TERMINATE --provisioning-model=STANDARD \
96         --image-family=ubuntu-2404-lts-amd64 --image-project=ubuntu-os-cloud \
97         --boot-disk-size=120GB --boot-disk-type=pd-balanced \
98         --local-ssd=interface=NVME \
99         --service-account=rally-worker@gen-lang-client-0306026826.iam.gserviceaccount.com \
100         --scopes=cloud-platform
101     ```
102
103     **Spot variant (cheaper; still needs the $2 quota ? Spot is NOT a quota bypass):** same
command with
104     `--provisioning-model=SPOT` (replacing `STANDARD`). Add `--instance-termination-
action=STOP` to keep
105     the disk on preemption so a re-run resumes. Cheaper, can be preempted.
106
107     > **Validated image (2026-06-20):** `--image-family=common-cu129-ubuntu-2204-nvidia-580
108     > --image-project=deeplearning-platform-release` ships the NVIDIA driver (no install-on-
boot wait;
109     > `nvidia-smi` works immediately) ? used for the first real L4 run. `gpu_setup.sh`
installs miniconda
110     > on it. **T4 was capacity-stocked-out in `asia-south1-a`; an L4 (`g2-standard-4`, no
`--accelerator`
111     > flag ? L4 is built into g2) in `asia-south1-c` landed** ? try L4/other zones if T4 is
unavailable.
112
113     ### 3b. Observability ? Ops Agent (GPU graphs + logs in the console)
114     The VM **Observability ? GPU** graphs + Logs need the **Ops Agent** (the console's OS-
policy install is
115     blocked on DLVM images). Install it directly + grant the worker SA the write roles (it
otherwise only has
116     storage), then GPU/NVML metrics + syslog flow to Cloud Monitoring/Logging (tab populates
in ~2-3 min):
117     ```bash
118     P=gen-lang-client-0306026826; SA=rally-worker@$P.iam.gserviceaccount.com
119     # 1. ENABLE the APIs (bare Gemini project has them OFF ? the agent installs but its
exports get
120     # PermissionDenied/SERVICE_DISABLED until these are on; THIS was the empty-graphs
cause 2026-06-20):
121     gcloud services enable logging.googleapis.com monitoring.googleapis.com --project $P
122     # 2. let rally-worker publish metrics + logs:
123     gcloud projects add-iam-policy-binding $P --member="serviceAccount:$SA"
--role=roles/monitoring.metricWriter --condition=None
124     gcloud projects add-iam-policy-binding $P --member="serviceAccount:$SA"
--role=roles/logging.logWriter --condition=None
125     # 3. on the box (created with --scopes=cloud-platform):
126     curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh && sudo

```

```

bash add-google-cloud-ops-agent-repo.sh --also-install
127     # (if the agent was installed BEFORE the APIs were enabled: `sudo systemctl restart
google-cloud-ops-agent`)
128     ```
129     > Best: bake `curl ?add-google-cloud-ops-agent-repo.sh|bash --also-install` into the VM
130     > `--metadata=startup-script` so observability is automatic on every box.
131
132     > **Long remote steps must be detached.** A plain `ssh ? "<long cmd>"` dies if the SSH
drops (seen
133     > mid-build, 2026-06-20) ? run them `nohup`'d on the box (`nohup bash setup.sh >log 2>&1
&`) and poll
134     > the log, so a dropped connection can't kill the install.

```

25. assistant (2026-06-21T09:25:55.181Z)

I now have everything confirmed. The Ops-Agent codify is clearly Claude-doable: the 3 pieces (enable logging/monitoring APIs in `provision.sh`; add the 2 SA roles in `provision.sh`; bake the agent install into a `--metadata=startup-script` in the README §3 VM-create command / a helper script) are all documented but not codified. The README even self-notes "Best: bake ? into the VM startup-script" (line 129).

Let me verify the BACKLOG entry for the DO token/droplet and confirm the repo rename scope, then check the COMPLETED stubs (finding #4) state quickly.

26. user (2026-06-21T09:25:55.796Z)

```

1     # Backlog ? deferred items to revisit
2
3     Single tracker for work we consciously deferred, so nothing gets lost between sessions.
Each item records **what**, **why it's deferred**, and **the trigger to revisit**. Every entry
traces to a real decision/PR/doc (no speculative work).
4
5     > Maintained as a version-controlled doc for now. Say the word and I'll mirror any/all
of these to **GitHub issues** instead (or in addition).
6     > Status legend: ? blocks something · ? do when triggered · ? nice-to-have/cleanup.
7
8     _Last updated: 2026-06-10._
9
10    ---
11
12    ## Owned-model / multisport (source: `OWNED_MODEL_IMPLEMENTATION_PLAN.md`,
`NEXT_STEPS.md`)
13    - ? **Rename the repo off "badminton" ? owner-emphatic "ASAP" action; the project is
multisport. Touches remote name + README/CLAUDE.md framing + any hardcoded URLs. **Revisit:**
owner action, before broad external sharing.
14    - ? **Quantify cost-to-Gen-2 (the largest open risk) ? benchmark real VRAM/token/$ for
the VLM SFT+GRPO on a rented A100/L40S (ms-swift multimodal-GRPO examples use 6?8 GPUs)
**before** standing up any owned serving stack. **Revisit:** before Phase-2 commitment.
15    - ? **Define the ship-gate F1@tIoU target** ? "very high P/R" must be operationalized
(floor = beat heuristic LOVO 0.480; learned is now 0.626 post metric-fix). **Revisit:** at M0.4
harness.
16
17    ## Engineering hardening deferred from the 2026-06-10 audit sweep
18    - ? **App-factory / lifespan + lazy torch-cv2 imports** ? `uvicorn backend.main:app`
leaves globals None, and importing the API pulls the GPU stack. Belongs with the async-job slice
(#16). **Revisit:** when the async slice starts.
19    - ? **WSL temp/frame/cache + Gemini remote-file GC** ? staged videos, PNG frame dumps,
`*_wasbcache`, and orphaned Gemini uploads are never cleaned. Needs a retention policy.
**Revisit:** before any volume/batch processing.
20    - ? **shlex-quote WSL config values** ? **DONE 2026-06-11 (#97):**
`detectors/base.py:wsl_tilde_quote` quotes everything after a leading `~/`, so quoting and
tilde expansion coexist; applied to every `~`-capable arg in both runners;
`tests/test_wsl_tilde_quoting.py` pins the command shapes (no quoted tilde can ever reach
`_wsl_bash`). Found live: the naive quoting had silently broken wasb_hybrid staging for the
default `~/clips`. Residual for tenancy-era: quote the remaining trusted-config interpolations

```

```

(repo_dir/conda_env) if config ever becomes per-tenant.
21
22     ---
23
24     ## Monetization / licensing
25     *(source: `archives/MONETIZATION_AUDIT.md` $7, ADR-005)*
26
27     - ? **Ultralytics Enterprise pricing** ? get a quote **only if** the
torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently **not
needed** (swap done). Quote-only pricing. **Revisit:** if detector quality forces reconsidering
YOLOv8.
28     - ? **Codec patent thresholds (AVC/HEVC/AAC)** ? confirm exact Via-LA (AVC) / Access
Advance (HEVC) royalty rates & free thresholds **with legal counsel** before relying on the
"small SaaS is under threshold" reading. Mitigation already chosen: emit royalty-free AV1/Opus
output. **Revisit:** before commercial launch.
29     - ? **Per-checkpoint weight/dataset licenses** ? if we borrow pretrained weights (YOLOX,
ActionFormer, WASB, TrackNetV3), the **code** license ? the distributed **weights/dataset** terms.
Either train our own or confirm each checkpoint. **Revisit:** when adopting any borrowed
weights.
30     - ? **Real GPU $/video & latency** ? benchmark the three LLM-step options (offline
ActionFormer vs self-hosted Qwen2.5-VL-7B vs Gemini paid-tier) on the target VM before pricing
the product. **Revisit:** when choosing the production LLM step.
31
32     ## Detector swap follow-ups
33     *(source: PR #8, ADR-005)*
34
35     - ? **`supervision` ByteTrack deprecation** ? **`sv.ByteTrack` is deprecated as of
supervision 0.28.0 and **removed in 0.30.0**; **`requirements.txt` is pinned **`<0.30.0` as a
stopgap. Migrate to **`ByteTrackTracker` from the separate **`trackers` package before lifting the
cap. **Revisit:** before bumping supervision past 0.30.
36     - ? **Real-video validation of the swap** ? the torchvision + ByteTrack Stage-1 is unit-
tested with mocks but **not yet validated on actual footage** for detection/tracking quality.
(ADR-004 separately found player pixel-motion is a weak rally proxy on broadcast video, so
expect tuning.) **Revisit:** when a real test clip + the pipeline are run together.
37     - ? **`yolo_hybrid` naming** ? the registry key, **`yolo_` config keys, and **`yolo-
hybrid-` DB detector tags keep the "yolo" name for back-compat despite no YOLO remaining.
Optional rename. **Revisit:** if/when a clean break is worth the migration.
38
39     ## Golden-set subproject
40     *(source: `docs/GOLDEN_SET_IMPLEMENTATION.md`, `docs/GOLDEN_SET_VENDORING.md`)*
41
42     - ? **Annotation guideline** ? **DONE (2026-06-02):** authored as the Roora brief +
illustrated guide in the collector repo **`docs/annotation/` ? rally CSV schema; ignore dead time
/ include every rally (incl. faults & lets) / shot = one contact; **`ending_reason`
(forced/unforced-error taxonomy, shared across net-separated sports); singles & doubles for
badminton & pickleball.
43     - ? **Internal gold key** ? hand-annotate 5?10 licensed clips to near-perfect as the
vendor answer key. **Unblocked.** **Revisit:** anytime.
44     - ? **GS-3 ? `HumanVendorProvider`** ? concrete adapter for the chosen annotation vendor
(submit/poll/fetch + DPA/secure-link controls). **Vendor selected for the initial pilot: Roora**
(Bengaluru; iMerit / Taskmonk / TELUS remain the fallback shortlist). **Revisit:** once the
Roora pilot confirms quality, then build the adapter.
45     - ? **GS-4 oracle model** ? **AutomatedProvider` interface is stubbed (this work) with a
deterministic fake. Wire a **concrete* fast-tier auto-labeler later. **Oracle rule:** it MUST be
a different model lineage than production, or the regression test is blind to shared failures.
**Revisit:** when a production LLM step is chosen (pick the oracle's lineage to differ).
46     - ? **GS-5 ? regression triggers** ? wire **`regress()` into CI (pre-merge gate), on-
demand CLI (the "surprise" case), and a scheduled run; include a deliberate re-baseline path.
**Revisit:** after GS-4.
47     - ? **Vendor pilot (Roora)** ? run the paid pilot with **Roora** as a **single-vendor
pilot first** (not a multi-vendor bake-off): assess quality-per-rupee against the $4 bar;
escalate to the iMerit/Taskmonk fallback only if it misses. **Revisit:** the pilot doubles as
calibration once the gold key exists.
48     - ? **Phase-1 video collection** ? gather the round-1 golden corpus per
`docs/GOLDEN_SET_PHASE1_VIDEOS.md`: **static self-recorded** footage (mirror deployment, not

```

broadcast), 3 sports incl. singles & doubles, across the diversity matrix; ~12 clips/sport with
 ? held out by match. **Unblocked.**

49 - ? **Train/eval separation guardrails** (was `DEFERRED\_HARDENING\_PLAN` #13; design
 preserved here on archival 2026-06-14) ? train & eval videos live in **separate catalogs**
 today, kept apart by convention only; nothing *enforces* that a video used to train the learned
 segmenter (`backend/eval/training.py::build\_training\_set` ? `distill\_local`) is absent from the
 eval/regression set, so leakage would silently inflate the headline metrics. **Concrete design**
 (from the archived DEFERRED plan §2): split unit = **match, never clip** (hold ~? of each sport
 as eval, per `GOLDEN\_SET\_PHASE1\_VIDEOS.md`); add a `split: "train"|"eval"` field to the
 collector's `curate export` manifest (cross-repo contract, ADR-002); **[DONE ? pure-logic core,
 PR #171]** `backend/eval/splits.py` now exists with `load_split(manifest) -> (train_ids,
 eval_ids)` keyed by **match id** (not file MD5 ? re-encodes share a match but differ in
 `compute\_video\_id`) + `assert\_disjoint(train\_ids, eval\_ids)` that raises on overlap, with
 `tests/test\_splits.py` (overlapping ids ? `assert\_disjoint` raises). **STILL DEFERRED:** wire
 the guard into the training entrypoint (`distill\_local`/`training`) and into `run\_regression.py`
 (baseline computed strictly on the eval split); add the collector's `curate export` manifest
 `split` field (above); the integration test (a deliberately-leaked manifest fails the training
 entrypoint). Full rationale/risks: [`archives/DEFERRED_HARDENING_PLAN-
 COMPLETED-2026-06-14.md`](archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md) §2.
 Revisit: dedicated design session with the user (needs a collector-side manifest change).

50

51 ## Recording best practices

52 *(source: `docs/VIDEO\_RECORDING\_GUIDELINES.md`)*

53

54 - ? **Validate recording hypotheses** ? empirically test which capture conditions
 (resolution, fps, camera angle/height, lighting) actually move precision/recall, then promote
 the winners into user-facing "how to record" guidance. Only the static-camera rule is currently
 evidence-backed (ADR-004). **Revisit:** once a controlled corpus + the eval loop can measure
 per-condition deltas.

55

27. assistant (2026-06-21T09:25:59.252Z)

I have all the grounding I need. Let me verify the COMPLETED-stub state for finding #4 (whether
 top-level `DEFERRED\_HARDENING\_PLAN.md` etc. exist as stubs vs only-archived) since I'll classify
 that doc-hygiene gate.

28. user (2026-06-21T09:26:00.401Z)

```
docs\CODE_AUDIT_AND_TEST_HARDENING.md
docs\DEFERRED_HARDENING_PLAN.md
docs\HARDENING_LOOP_HANDOFF.md
docs\HOSTING_PLAN.md
```

29. user (2026-06-21T09:26:02.103Z)

```
11 docs/DEFERRED_HARDENING_PLAN.md
13 docs/CODE_AUDIT_AND_TEST_HARDENING.md
15 docs/HARDENING_LOOP_HANDOFF.md
39 total
```

30. assistant (2026-06-21T09:26:12.810Z)

Confirmed: the three top-level docs are short pointer stubs (11-15 lines each), with archived
 full copies. Finding #4 is purely a confirm/document decision ? Claude-doable doc-hygiene.

I now have everything grounded. Let me assemble the structured output covering my area: cross-
 track gates + housekeeping outside the M-series.

Summary of what I verified:

- **P3 reconcile**: M3 (#282) added `backend/utils/paths.py::output\_base(config)` and rerooted
 the chunk dirs under it (`trajectory\_hybrid.py:314`, `yolo\_hybrid.py:269,408`, `gemini.py:540`),
 but the **folder names** `chunks\_{provider}\_handoff` and `chunks\_yolo` are still hardcoded. So
 M3 **partially** subsumed P3 (kills the hardcoded `output/` base) but **not fully** (the per-

video `tmp/` convergence + workspace routing is still P3's job). The reconcile = a small doc PR.

- **Ops-Agent codify**: `deploy/gcp/provision.sh` does storage+IAM+budget only; logging/monitoring APIs + the 2 SA roles + agent install are documented in README §3b (lines 119-129) but NOT codified. README line 129 itself says "bake into the VM startup-script". Claude-doable.
- **DO droplet/token, repo rename, WASB-C3, collector md5**: all owner/counsel/cross-repo actions, grounded in DECISIONS.md:681, BACKLOG.md:13, COMMERCIALIZATION.md:65, and the collector-repo `curate export`.
- **Doc-hygiene leftovers** (DOC_STATUS §4 #4/#5, NEXT_STEPS line 31): Claude-doable doc PRs.

31. user (2026-06-21T09:27:39.615Z)

Structured output provided successfully

32. assistant (2026-06-21T09:27:53.743Z)

I have mapped all cross-track gates and housekeeping items in my area. Summary of the key findings:

Claude-doable NOW (no owner needed) ? 5 housekeeping PRs:

1. **P3 reconcile** ? VERIFIED: M3 (#282) added `backend/utils/paths.py::output\_base()` and rerooted chunk dirs under it, but `chunks\_yolo` / `chunks\_{provider}\_handoff` folder names are STILL hardcoded literals (`yolo\_hybrid.py:408`, `trajectory\_hybrid.py:314`, `gemini.py:540`). So M3 subsumed only the *output-base* half of P3, not the per-video `tmp/` routing. A doc-only PR closes the reconcile and fixes stale `:237,:313` anchors ? now `:314`.
2. **Ops-Agent codify** ? VERIFIED: `provision.sh` does storage+IAM+budget only (no logging/monitoring APIs, only `storage.objectAdmin`); the 3 Ops-Agent pieces live solely as README §3b prose (`README.md:119-129`), which itself says "bake into the VM startup-script". Fully Claude-authorable infra PR; owner just runs it next provision.
- 3-5. Three doc-hygiene PRs (COMPLETED-stub confirm, HOSTING_PLAN banner, finish the already-mostly-done archive relocation).

Owner-prerequisite gates (nothing Claude-doable in this repo until the owner acts):

- **DO droplet retire** (run_execution, ~\$24/mo) ? **DO token retire** (creds, depends on droplet gone)
- **Repo rename off "badminton"** (design_choice ? owner must pick the name + rename the remote; then Claude can sweep in-repo URLs/prose)
- **WASB-C3** (counsel_consent ? tracking-only, explicitly "no code change in this repo until resolved upstream")
- **Collector md5 export** (design_choice ? sibling-repo change; the indexer consumer `backend/eval/splits.py` is already built per PR #171)

The highest unblock-value owner decisions in my area are the **repo rename** (frees an in-repo URL/prose sweep PR) ? everything else owner-gated is standing-cost cleanup or distant licensing/cross-repo work that gates the *serving/sharing* step, not current M5+ engine PRs.